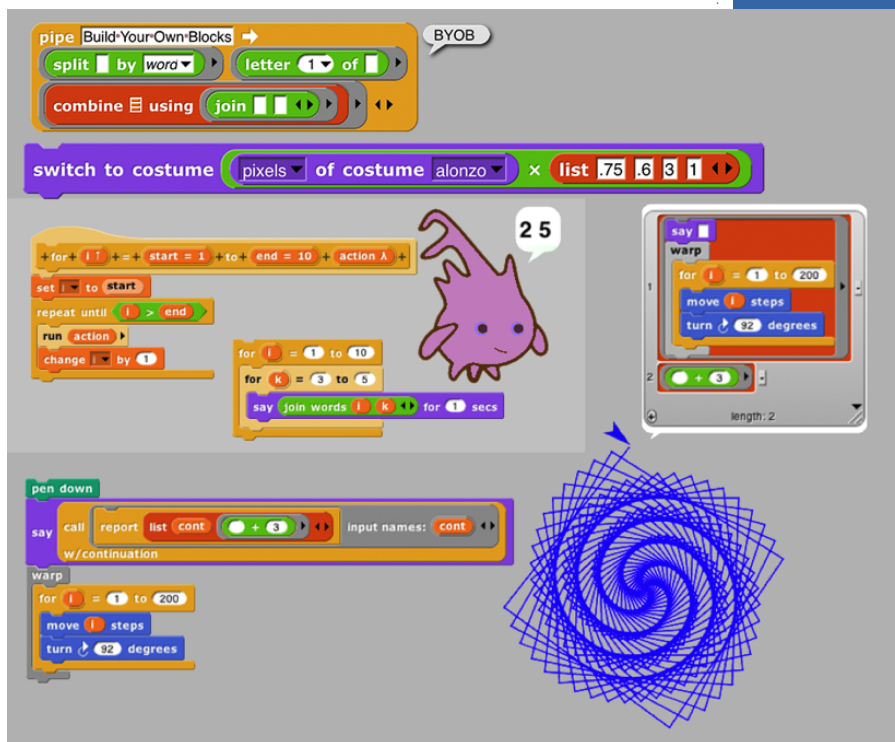




Build Your Own Blocks

8.0

SNAP! REFERENCE MANUAL



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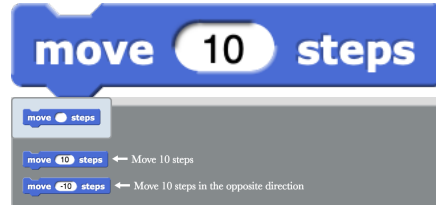
0.1 All Snap! Blocks

Block	Text
	move
	turn
	_ deg
	turn
	\$count
	_ deg
	point
	dire
	point
	go to
	go to
	glide
	x: _
	chang
	set :
	chang
	set :
	if on
	boun
	posi
	x pos
	y pos
	dire
	swit
	cost
	next
	cost
	say .
	say .
	thin
	secs
	thin
	_ of
	stre
	y: _
	skew
	degr
	new c
	widt
	chang
	by

0.2 Motion Blocks

0.3 move steps

Moves a sprite a set number of steps (one coordinate) in whichever direction the sprite is facing.



0.3.1 Example Images

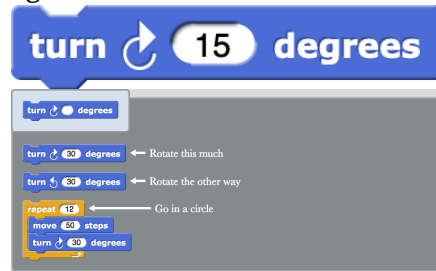
Create a square shape

0.3.2 Example Projects

- Move Steps Example 1
- Move Steps Example 2
- Move Steps Example 3

0.4 turn ⌚ degrees

Turns the sprite a specified degree of angle in the clockwise direction



0.4.1 Example Images

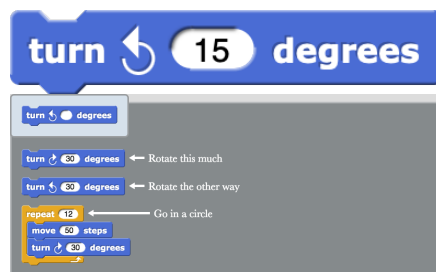
No examples yet.

0.4.2 Example Projects

No examples yet.

0.5 turn 15 degrees

Complete Me



0.5.1 Example Images

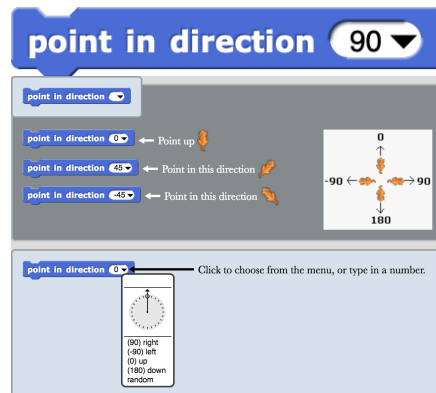
No examples yet.

0.5.2 Example Projects

No examples yet.

0.6 point in direction

Complete Me



0.6.1 Example Images

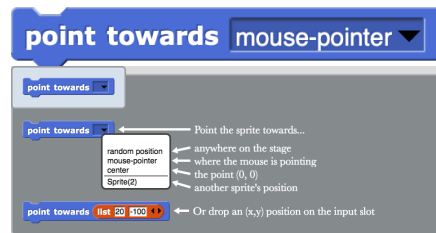
No examples yet.

0.6.2 Example Projects

No examples yet.

0.7 point towards

Complete Me



0.7.1 Example Images

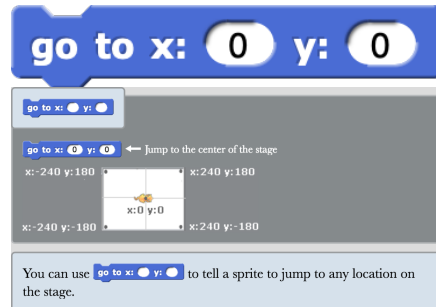
No examples yet.

0.7.2 Example Projects

No examples yet.

0.8 go to x: y:

Complete Me



0.8.1 Example Images

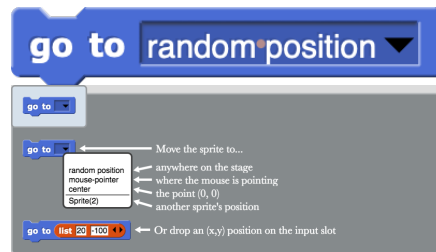
No examples yet.

0.8.2 Example Projects

No examples yet.

0.9 go to

Complete Me



0.9.1 Example Images

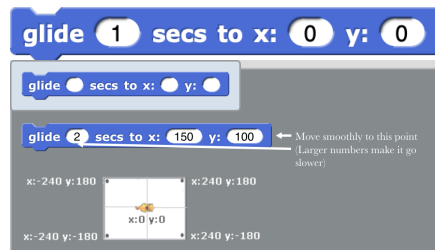
No examples yet.

0.9.2 Example Projects

No examples yet.

0.10 Glide Secs to Position

Complete Me



0.10.1 Example Images

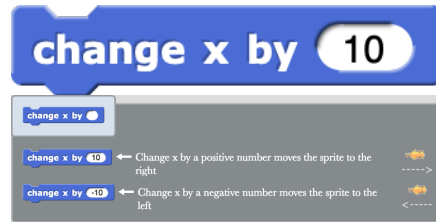
No examples yet.

0.10.2 Example Projects

No examples yet.

0.11 change x by

Complete Me



0.11.1 Example Images

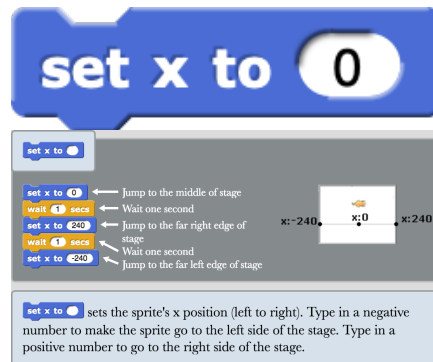
No examples yet.

0.11.2 Example Projects

No examples yet.

0.12 set x to

Complete Me



0.12.1 Example Images

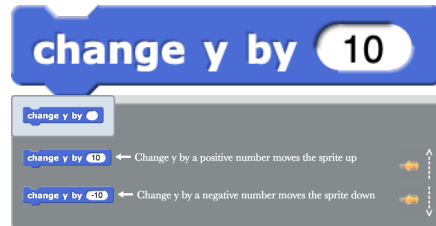
No examples yet.

0.12.2 Example Projects

No examples yet.

0.13 change y by

Complete Me



0.13.1 Example Images

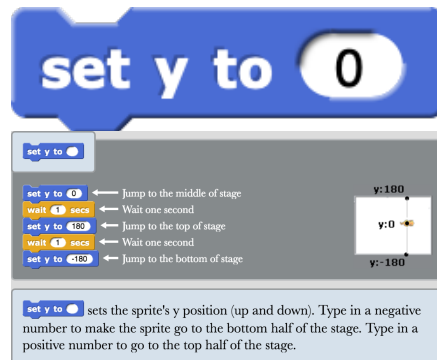
No examples yet.

0.13.2 Example Projects

No examples yet.

0.14 set y to

Complete Me



0.14.1 Example Images

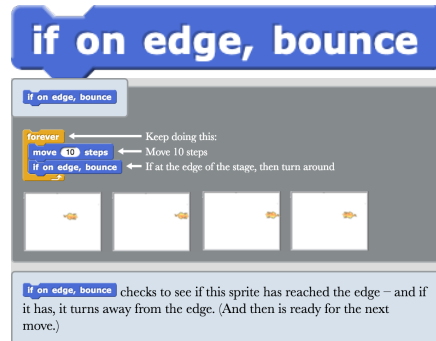
No examples yet.

0.14.2 Example Projects

No examples yet.

0.15 if on edge, bounce

Complete Me



0.15.1 Example Images

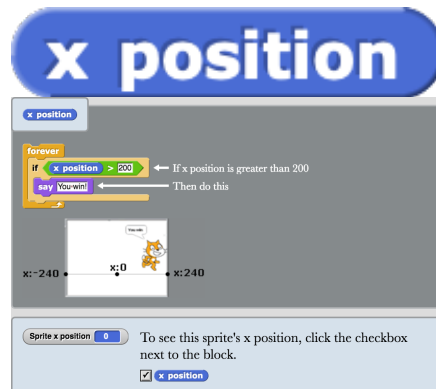
No examples yet.

0.15.2 Example Projects

No examples yet.

0.17 x position

Complete Me



0.17.1 Example Images

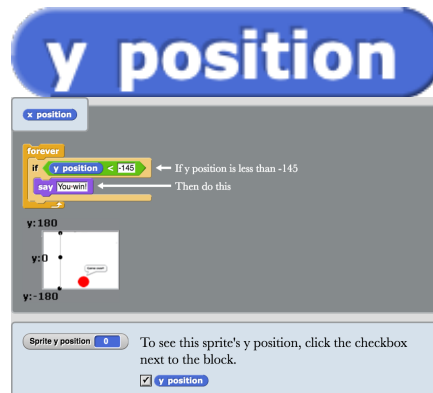
No examples yet.

0.17.2 Example Projects

No examples yet.

0.18 y position

Complete Me



0.18.1 Example Images

No examples yet.

0.18.2 Example Projects

No examples yet.

0.19 direction

Complete Me



0.19.1 Example Images

No examples yet.

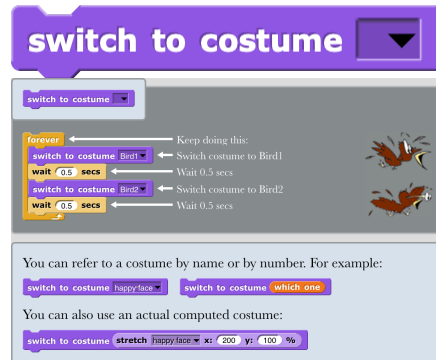
0.19.2 Example Projects

No examples yet.

0.20 Looks Blocks

0.21 switch to costume

Complete Me



0.21.1 Example Images

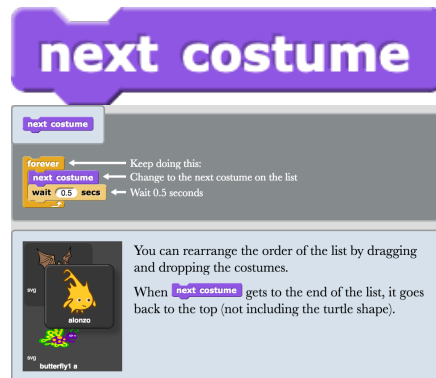
No examples yet.

0.21.2 Example Projects

No examples yet.

0.22 next costume

Complete Me



0.22.1 Example Images

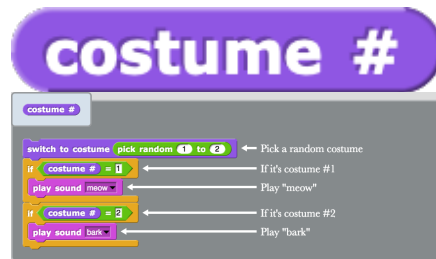
No examples yet.

0.22.2 Example Projects

No examples yet.

0.23 costume number

Complete Me



0.23.1 Example Images

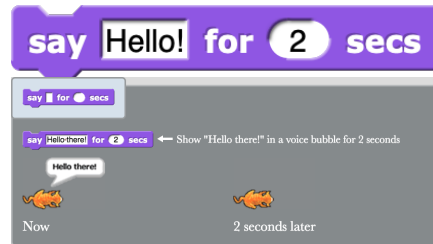
No examples yet.

0.23.2 Example Projects

No examples yet.

0.24 say for secs

Complete Me



0.24.1 Example Images

No examples yet.

0.24.2 Example Projects

No examples yet.

0.25 say

Complete Me



0.25.1 Example Images

No examples yet.

0.25.2 Example Projects

No examples yet.

0.26 think for secs

Complete Me



0.26.1 Example Images

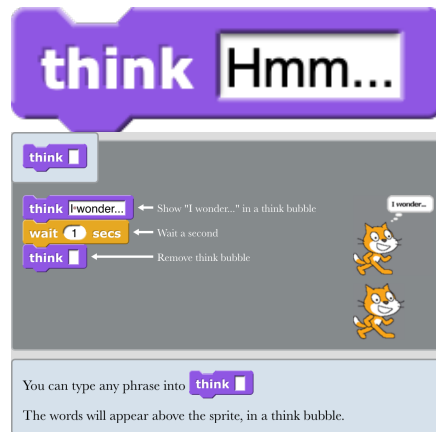
No examples yet.

0.26.2 Example Projects

No examples yet.

0.27 think

Complete Me



0.27.1 Example Images

No examples yet.

0.27.2 Example Projects

No examples yet.

0.28 Attribute of Costume

Complete Me

width of costume current

width of costume

width of costume

width
height
pixels

Turtle
current
wings
ballerina
beard
blue

row	A	B	G	R
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	45	7	200	200
9	45	7	200	200
10	45	8	200	200
11	45	8	200	200
12	45	8	200	200
13	45	8	200	200
14	45	8	200	200
15	45	8	200	200
16	45	8	200	200
17	45	8	200	200
18	45	8	200	200
19	45	8	200	200
20	45	8	200	200

pixels of costume costume

Report attributes of a costume.

Width and height are reported in pixels, but don't change in presentation mode. Pixels are reported as a list of lists, in which each sublist is the RGBA values for one pixel (Red, Green, Blue, Alpha). "Alpha" is the opacity (inverse of ghost effect). All values are between 0 and 255.

The pixel list does not have width or height information; it needs a width to make it a viewable costume. [show picture](#) in the Pixels library uses the current costume's width.

0.28.1 Example Images

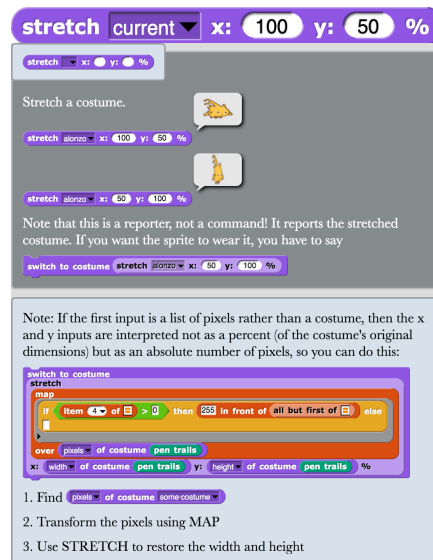
No examples yet.

0.28.2 Example Projects

No examples yet.

0.29 Stretch Costume

Complete Me



The image shows a Scratch 'stretch' block with a dropdown menu set to 'current'. The 'x' input is 100 and the 'y' input is 50, followed by a '%' symbol. Below the block, there is a script area with two 'stretch' blocks. The first block has 'stretch' set to 'along', 'x' set to 100, and 'y' set to 50. The second block has 'stretch' set to 'along', 'x' set to 50, and 'y' set to 100. A note explains that the 'stretch' block is a reporter, not a command, and that it reports the stretched costume. It also provides a 'switch to costume' block that uses the 'stretch' block. A detailed note explains that if the first input is a list of pixels, the 'x' and 'y' inputs are interpreted as absolute numbers of pixels. An example script is shown: a 'switch to costume' block followed by a 'stretch' block. The 'stretch' block has a 'map' block inside it. The 'map' block has a condition 'if item 1 of <list of pixels> > 0 then 255 in front of all but first of <list of pixels> else 1'. The 'map' block is followed by an 'over' block with inputs 'pixels' of costume, 'pen trails', 'x' width of costume, 'pen trails', 'y' height of costume, 'pen trails', and '%'. A list of steps is provided: 1. Find pixels of costume some costume, 2. Transform the pixels using MAP, 3. Use STRETCH to restore the width and height.

stretch current x: 100 y: 50 %

stretch x: 100 y: 50 %

stretch x: 50 y: 100 %

Note that this is a reporter, not a command! It reports the stretched costume. If you want the sprite to wear it, you have to say

switch to costume stretch x: 50 y: 100 %

Note: If the first input is a list of pixels rather than a costume, then the x and y inputs are interpreted not as a percent (of the costume's original dimensions) but as an absolute number of pixels, so you can do this:

switch to costume stretch

map

if item 1 of <list of pixels> > 0 then 255 in front of all but first of <list of pixels> else 1

over pixels of costume pen trails x: width of costume pen trails y: height of costume pen trails %

1. Find pixels of costume some costume
2. Transform the pixels using MAP
3. Use STRETCH to restore the width and height

0.29.1 Example Images

No examples yet.

0.29.2 Example Projects

No examples yet.

0.30 Skew Costume by Degrees

Complete Me

skew current to 0 degrees 50 %

0.30.1 Example Images

No examples yet.

0.30.2 Example Projects

No examples yet.

0.31 New Costume

Complete Me



0.31.1 Example Images

No examples yet.

0.31.2 Example Projects

No examples yet.

0.32 change effect by

Complete Me



0.32.1 Example Images

No examples yet.

0.32.2 Example Projects

No examples yet.

0.33 set effect to

Complete Me



0.33.1 Example Images

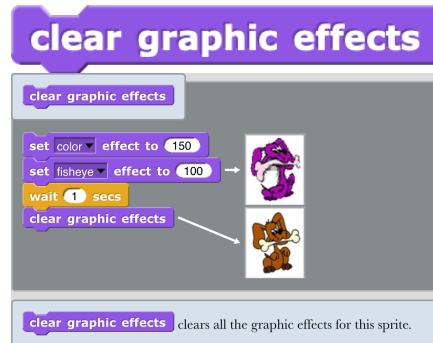
No examples yet.

0.33.2 Example Projects

No examples yet.

0.34 clear graphic effects

Complete Me



0.34.1 Example Images

No examples yet.

0.34.2 Example Projects

No examples yet.

0.35 _ effect

Complete Me



0.35.1 Example Images

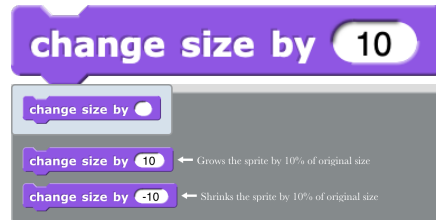
No examples yet.

0.35.2 Example Projects

No examples yet.

0.36 change size by

Complete Me



0.36.1 Example Images

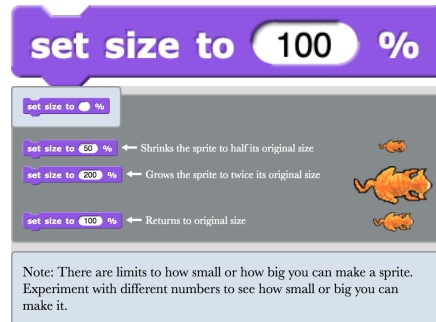
No examples yet.

0.36.2 Example Projects

No examples yet.

0.37 Set Sprite Size

Complete Me



0.37.1 Example Images

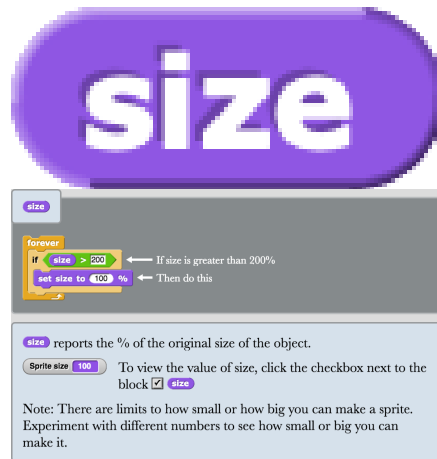
No examples yet.

0.37.2 Example Projects

No examples yet.

0.38 size

Complete Me



0.38.1 Example Images

No examples yet.

0.38.2 Example Projects

No examples yet.

0.39 **show**

Complete Me



0.39.1 **Example Images**

No examples yet.

0.39.2 **Example Projects**

No examples yet.

0.40 hide

Complete Me



0.40.1 Example Images

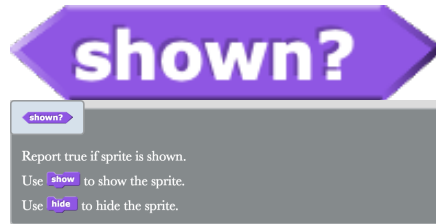
No examples yet.

0.40.2 Example Projects

No examples yet.

0.41 shown?

Complete Me



0.41.1 Example Images

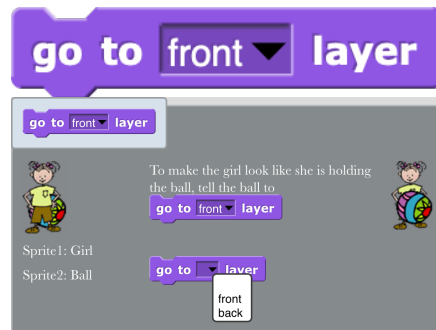
No examples yet.

0.41.2 Example Projects

No examples yet.

0.42 go to layer

Complete Me



0.42.1 Example Images

No examples yet.

0.42.2 Example Projects

No examples yet.

0.43 go back layers

Complete Me



0.43.1 Example Images

No examples yet.

0.43.2 Example Projects

No examples yet.

0.44 Sound Blocks

0.45 Play Sound

Complete Me



0.45.1 Example Images

No examples yet.

0.45.2 Example Projects

No examples yet.

0.46 Play Sound Until Done

Complete Me



0.46.1 Example Images

No examples yet.

0.46.2 Example Projects

No examples yet.

0.47 Stop All Sounds

Complete Me



0.47.1 Example Images

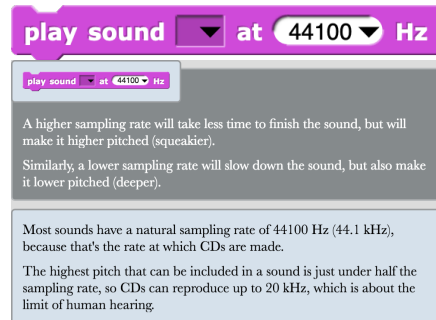
No examples yet.

0.47.2 Example Projects

No examples yet.

0.48 Play Sound Hz

Complete Me



0.48.1 Example Images

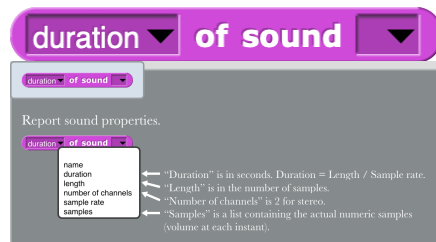
No examples yet.

0.48.2 Example Projects

No examples yet.

0.49 Attribute Of Sound

Complete Me



0.49.1 Example Images

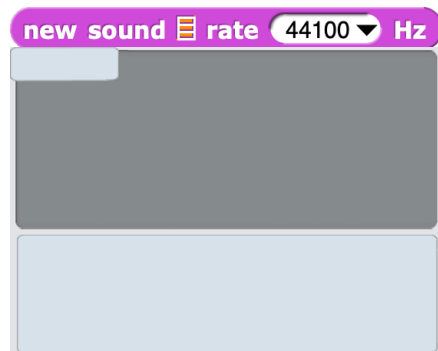
No examples yet.

0.49.2 Example Projects

No examples yet.

0.50 New Sound Rate Hz

Complete Me



0.50.1 Example Images

No examples yet.

0.50.2 Example Projects

No examples yet.

0.51 Rest For Beats

Complete Me



0.51.1 Example Images

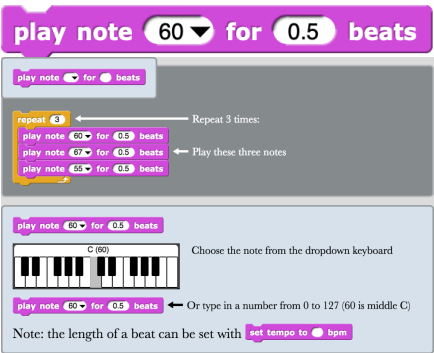
No examples yet.

0.51.2 Example Projects

No examples yet.

0.52 Play Note For Beats

Complete Me



0.52.1 Example Images

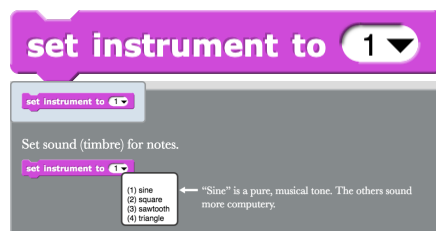
No examples yet.

0.52.2 Example Projects

No examples yet.

0.53 Set Instrument

Complete Me



0.53.1 Example Images

No examples yet.

0.53.2 Example Projects

No examples yet.

0.54 Change Tempo

Complete Me

change tempo by20

change tempo by

set tempo to60 bpm

Set tempo to 60 beats per minute

forever

Keep doing this

play noteC4for 0.5 beats

Play a note for 0.5 beats

change tempo by20

Increase the tempo by 20 beats

if tempo > 500

If the tempo is greater than 500

stop this script

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play noteC4for 0.5 beats ← 0.5 beats = 0.5 seconds at 60 bpm

play noteC4for 0.2 beats ← 0.2 beats = 0.1 seconds at 120 bpm

0.54.1 Example Images

No examples yet.

0.54.2 Example Projects

No examples yet.

0.55 Set Tempo

Complete Me

set tempo to 60 bpm

set tempo to 60 bpm

forever

play note C4 for 0.5 beats

change tempo by 20

if tempo > 500

stop this script

Set tempo to 60 beats per minute

Keep doing this

Play a note for 0.5 beats

Increase the tempo by 20 beats

If the tempo is greater than 500

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play note C4 for 0.5 beats ← 0.5 beats = 0.5 seconds at 60 bpm

play note C4 for 0.2 beats ← 0.2 beats = 0.1 seconds at 120 bpm

0.55.1 Example Images

No examples yet.

0.55.2 Example Projects

No examples yet.

0.56 Tempo

Complete Me

tempo

set tempo to 60 bpm

Set tempo to 60 beats per minute

forever

Keep doing this:

play note 64 for 0.5 beats

Play a note for 0.5 beats

change tempo by 20

Increase the tempo by 20 beats

if tempo > 500

If the tempo is greater than 500

stop this script

Stop this script

Tempo is the speed (bpm = beats per minute) at which Snap! notes and drums play. The larger the tempo value, the faster the notes and drums will play.

play note 60 for 0.5 beats

0.5 beats = 0.5 seconds at 60 bpm

play note 65 for 0.2 beats

0.2 beats = 0.1 seconds at 120 bpm

0.56.1 Example Images

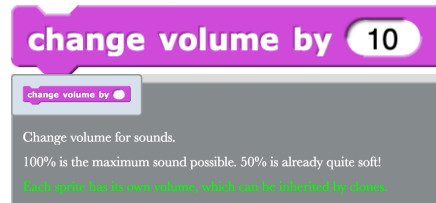
No examples yet.

0.56.2 Example Projects

No examples yet.

0.57 Change Volume

Complete Me



0.57.1 Example Images

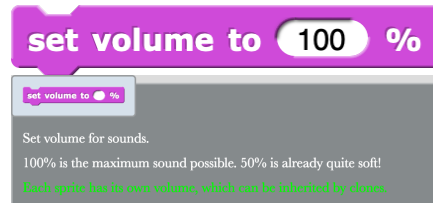
No examples yet.

0.57.2 Example Projects

No examples yet.

0.58 Set Volume

Complete Me



0.58.1 Example Images

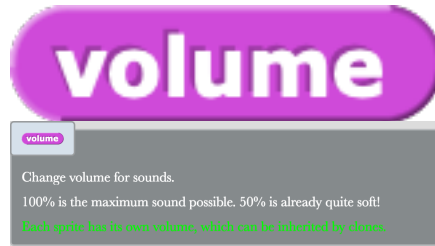
No examples yet.

0.58.2 Example Projects

No examples yet.

0.59 Volume

Complete Me



0.59.1 Example Images

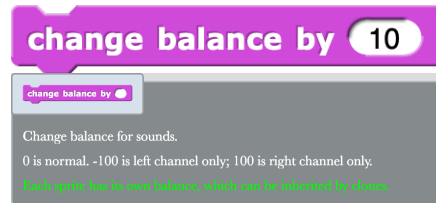
No examples yet.

0.59.2 Example Projects

No examples yet.

0.60 Change Balance

Complete Me



0.60.1 Example Images

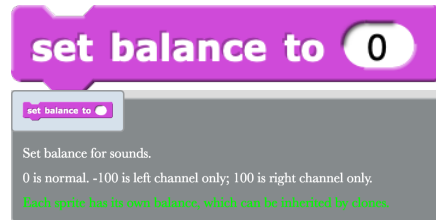
No examples yet.

0.60.2 Example Projects

No examples yet.

0.61 Set Balance To

Complete Me



0.61.1 Example Images

No examples yet.

0.61.2 Example Projects

No examples yet.

0.62 Report Balance

Complete Me



0.62.1 Example Images

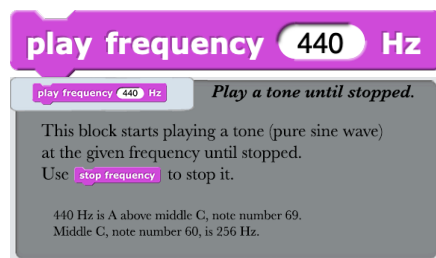
No examples yet.

0.62.2 Example Projects

No examples yet.

0.63 Play Frequency Hz

Complete Me



0.63.1 Example Images

No examples yet.

0.63.2 Example Projects

No examples yet.

0.64 Stop Frequency

Complete Me



0.64.1 Example Images

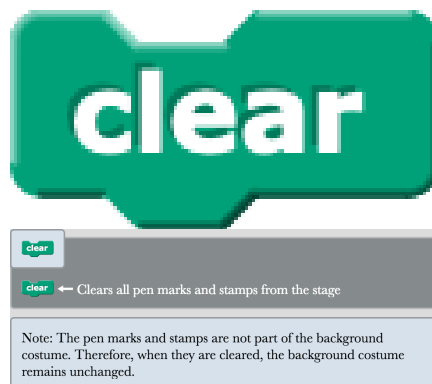
No examples yet.

0.64.2 Example Projects

No examples yet.

0.66 clear

Complete Me



0.66.1 Example Images

No examples yet.

0.66.2 Example Projects

No examples yet.

0.67 pen down

Complete Me



0.67.1 Example Images

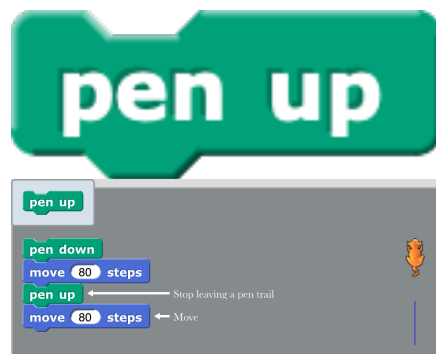
No examples yet.

0.67.2 Example Projects

No examples yet.

0.68 pen up

Complete Me



0.68.1 Example Images

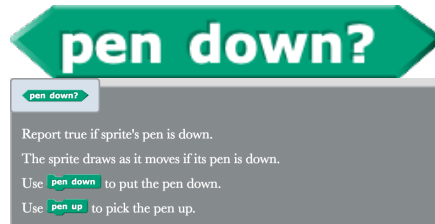
No examples yet.

0.68.2 Example Projects

No examples yet.

0.69 pen down?

Complete Me



0.69.1 Example Images

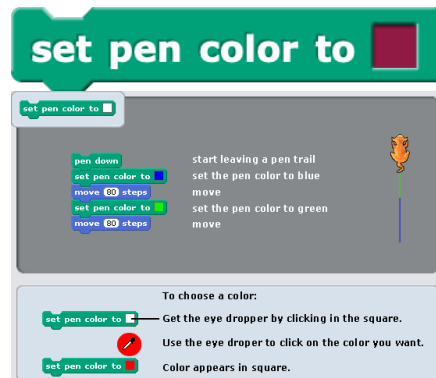
No examples yet.

0.69.2 Example Projects

No examples yet.

0.70 set pen color to

Complete Me



0.70.1 Example Images

No examples yet.

0.70.2 Example Projects

No examples yet.

0.71 change pen by

Complete Me



0.71.1 Example Images

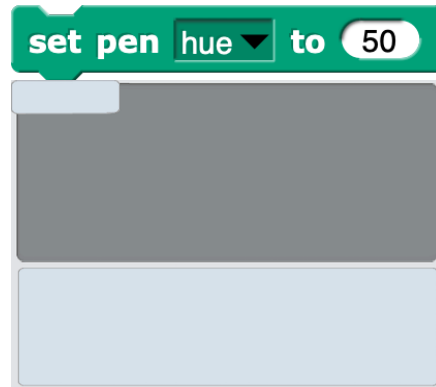
No examples yet.

0.71.2 Example Projects

No examples yet.

0.72 set pen to

Complete Me



0.72.1 Example Images

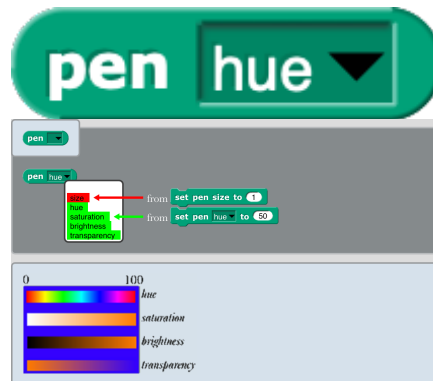
No examples yet.

0.72.2 Example Projects

No examples yet.

0.73 pen

Complete Me



0.73.1 Example Images

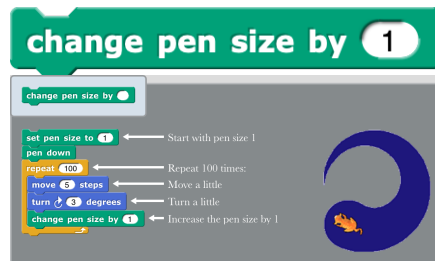
No examples yet.

0.73.2 Example Projects

No examples yet.

0.74 change pen size by

Complete Me



0.74.1 Example Images

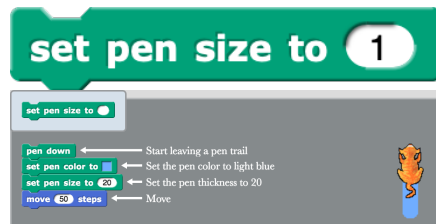
No examples yet.

0.74.2 Example Projects

No examples yet.

0.75 set pen size to

Complete Me



0.75.1 Example Images

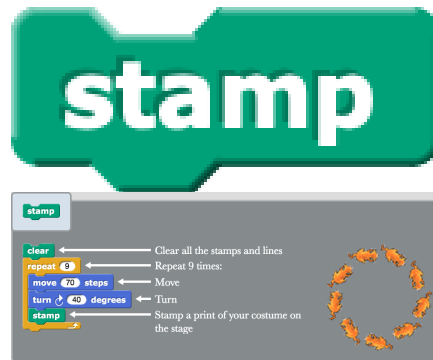
No examples yet.

0.75.2 Example Projects

No examples yet.

0.76 stamp

Complete Me



0.76.1 Example Images

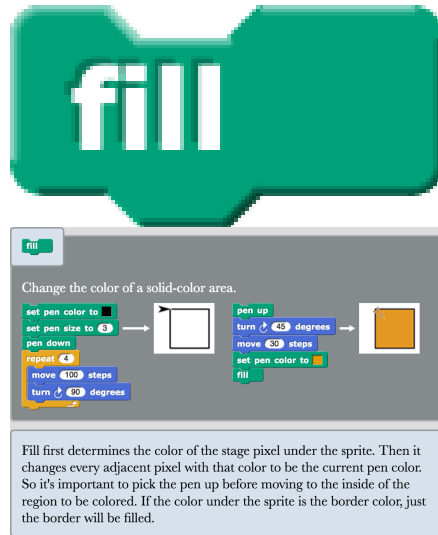
No examples yet.

0.76.2 Example Projects

No examples yet.

0.77 fill

Complete Me



0.77.1 Example Images

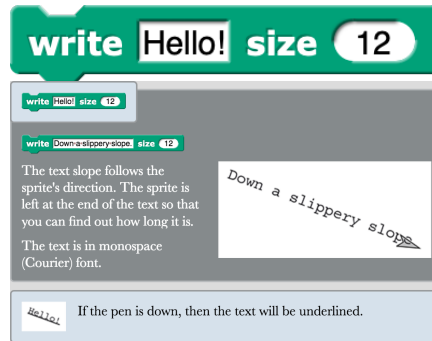
No examples yet.

0.77.2 Example Projects

No examples yet.

0.78 write size

Complete Me



0.78.1 Example Images

No examples yet.

0.78.2 Example Projects

No examples yet.

0.79 pen trails

Complete Me

pen trails

pen trails

Report lines drawn on stage as a costume.



pen trails

The reported value does not include the stage background or the sprite itself, just the lines drawn with the pen.

What can you do with the reported costume?

Add it to your costumes.

add pen trails to my costumes

Modify it. 

switch to costume

stretch

map

if item of then > 0 then 200 in front of all but first of else

over pixels of costume pen trails

x1 width of costume pen trails

y1 height of costume pen trails

%

0.79.1 Example Images

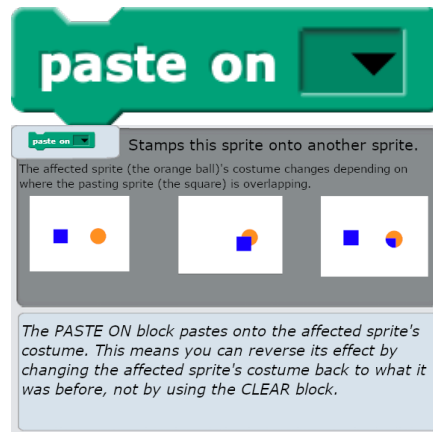
No examples yet.

0.79.2 Example Projects

No examples yet.

0.80 paste on

Complete Me



0.80.1 Example Images

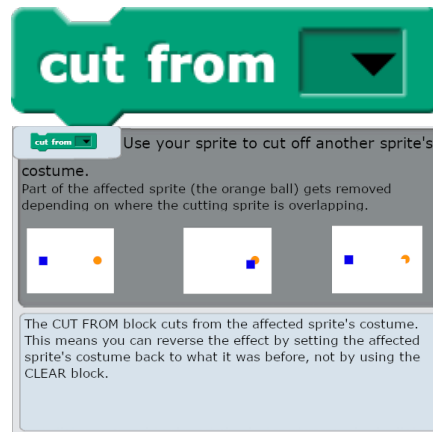
No examples yet.

0.80.2 Example Projects

No examples yet.

0.81 cut from

Complete Me



0.81.1 Example Images

No examples yet.

0.81.2 Example Projects

No examples yet.

0.82 Color

Complete Me



0.82.1 Example Images

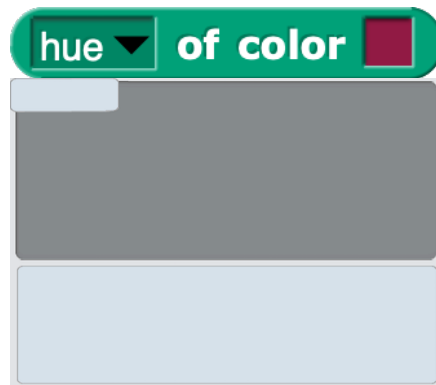
No examples yet.

0.82.2 Example Projects

No examples yet.

0.83 Color Attribute

Complete Me



0.83.1 Example Images

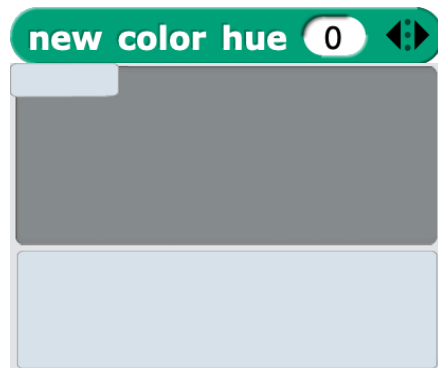
No examples yet.

0.83.2 Example Projects

No examples yet.

0.84 new color

Complete Me



0.84.1 Example Images

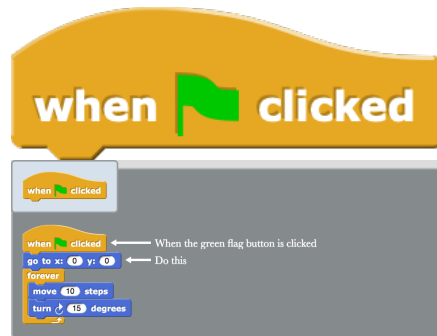
No examples yet.

0.84.2 Example Projects

No examples yet.

0.86 When Green Flag Clicked

Complete Me



0.86.1 Example Images

No examples yet.

0.86.2 Example Projects

No examples yet.

0.87 when is edited

Complete Me



0.87.1 Example Images

No examples yet.

0.87.2 Example Projects

No examples yet.

0.88 when I start as a clone

Complete Me

when I start as a clone

when I start as a clone

Clone initialization script

when clicked
repeat 3
create a clone of Sprite

→

(because the original and all three copies are in the same place and the same color)

when I start as a clone
set pen hue to pick random 1 to 100
set pen saturation to 100
set pen brightness to 100
go to x: pick random -50 to 50 y: pick random -50 to 50

when clicked
repeat 3
create a clone of Sprite

→

Note: Snap! clones are not copies of the sprite; they actually share information, so a change in the original may be seen in the clones. See the “Object Oriented Programming with Sprites” chapter in the Snap! Reference Manual.

0.88.1 Example Images

No examples yet.

0.88.2 Example Projects

No examples yet.

0.89 when I receive

Complete Me



0.89.1 Example Images

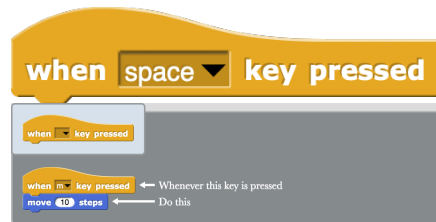
No examples yet.

0.89.2 Example Projects

No examples yet.

0.90 when key pressed

Complete Me



0.90.1 Example Images

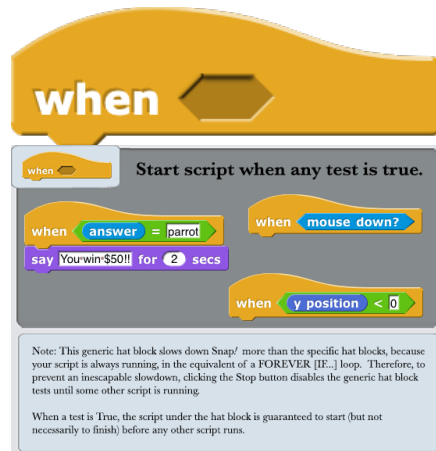
No examples yet.

0.90.2 Example Projects

No examples yet.

0.91 when

Complete Me



0.91.1 Example Images

No examples yet.

0.91.2 Example Projects

No examples yet.

0.92 when I am

Complete Me



0.92.1 Example Images

No examples yet.

0.92.2 Example Projects

No examples yet.

0.93 broadcast

Complete Me



0.93.1 Example Images

No examples yet.

0.93.2 Example Projects

No examples yet.

0.94 broadcast and wait

Complete Me



0.94.1 Example Images

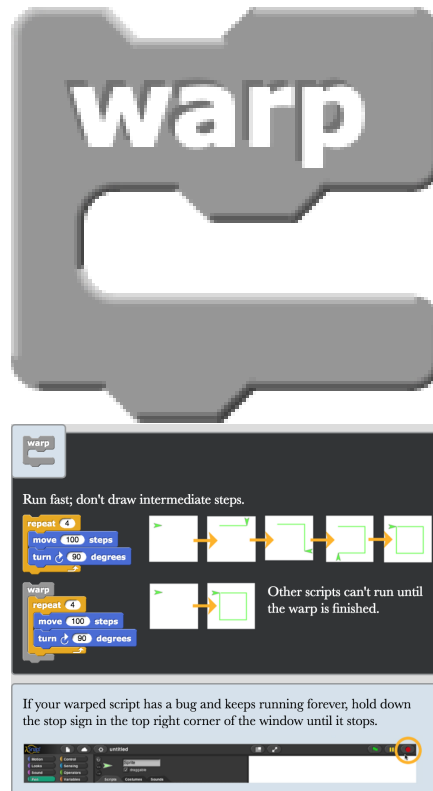
No examples yet.

0.94.2 Example Projects

No examples yet.

0.95 Warp

Complete Me



0.95.1 Example Images

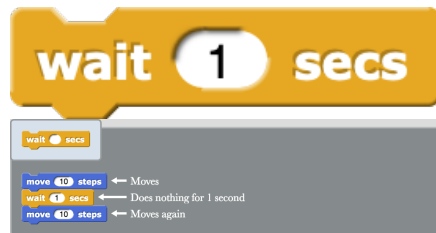
No examples yet.

0.95.2 Example Projects

No examples yet.

0.96 wait secs

Complete Me



0.96.1 Example Images

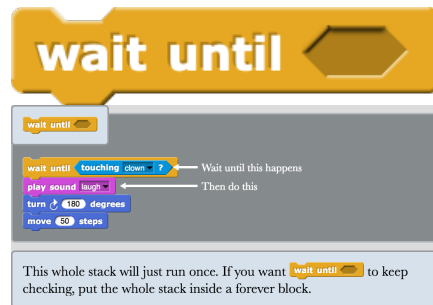
No examples yet.

0.96.2 Example Projects

No examples yet.

0.97 wait until

Complete Me



0.97.1 Example Images

No examples yet.

0.97.2 Example Projects

No examples yet.

0.98 forever

Runs the scripts in a loop until the stop button is clicked.



0.98.1 Example Images

No examples yet.

0.98.2 Example Projects

No examples yet.

0.99 repeat

Complete Me



0.99.1 Example Images

No examples yet.

0.99.2 Example Projects

No examples yet.

0.100 repeat until

Complete Me



0.100.1 Example Images

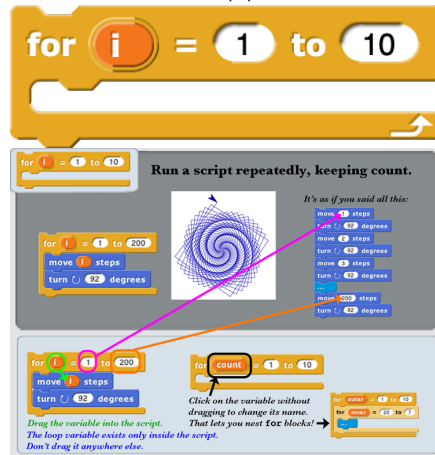
No examples yet.

0.100.2 Example Projects

No examples yet.

0.101 for _ = to

The for block lets you make a loop with a new variable (i) that starts a 1 and increments by 1 value, up to 10.



0.101.1 Example Images

No examples yet.

0.101.2 Example Projects

No examples yet.

0.102 if

Complete Me



0.102.1 Example Images

No examples yet.

0.102.2 Example Projects

No examples yet.

0.103 if else

Complete Me



0.103.1 Example Images

No examples yet.

0.103.2 Example Projects

No examples yet.

0.104 if then else

Complete Me



0.104.1 Example Images

No examples yet.

0.104.2 Example Projects

No examples yet.

0.105 report

Complete Me



0.105.1 Example Images

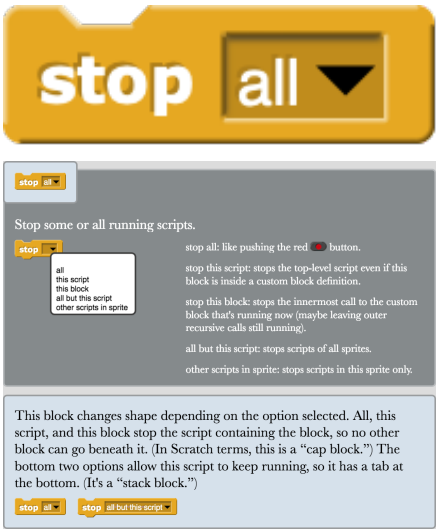
No examples yet.

0.105.2 Example Projects

No examples yet.

0.106 stop

Complete Me



0.106.1 Example Images

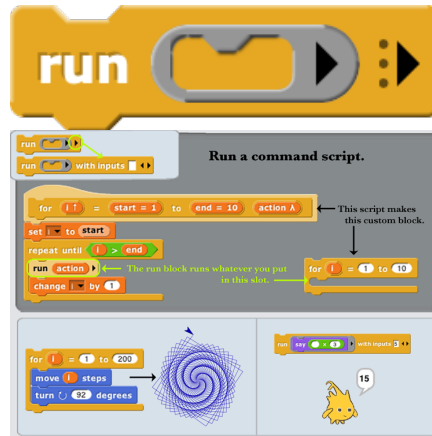
No examples yet.

0.106.2 Example Projects

No examples yet.

0.107 run

Complete Me



0.107.1 Example Images

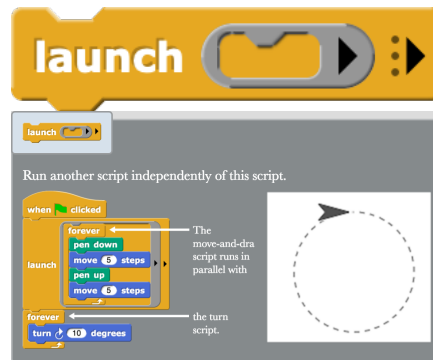
No examples yet.

0.107.2 Example Projects

No examples yet.

0.108 launch

Complete Me



0.108.1 Example Images

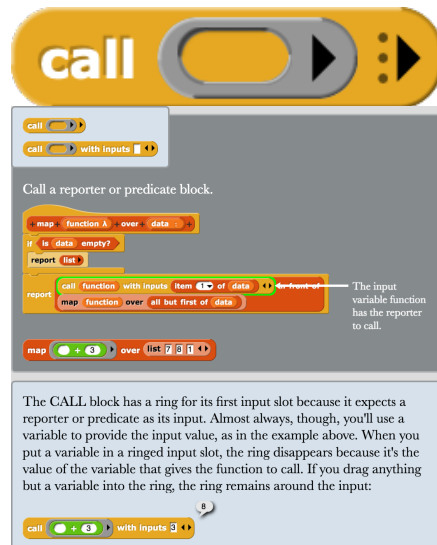
No examples yet.

0.108.2 Example Projects

No examples yet.

0.109 call

Complete Me



0.109.1 Example Images

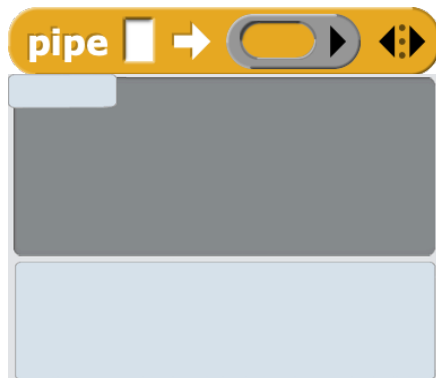
No examples yet.

0.109.2 Example Projects

No examples yet.

0.110 pipe →

Complete Me



0.110.1 Example Images

No examples yet.

0.110.2 Example Projects

No examples yet.

0.112 ask for

Complete Me

askfor

ask for

ask another object a question.
Ask a particular sprite where it is:

ask Sprite(3) for list x position y position

1	156
2	20

length: 2

ask all sprites except myself where they are:

2	A	B
1	-177	-76
2	156	20

map ask for list x position y position over my other sprite

(Each sprite is used, one at a time, to fill the empty slot circled in green.)

The expression is evaluated in the environment of the other object, so it uses that object's local variables and methods. If you want to use one of the sender's variables inside the script, leave an empty input slot in the script, click the right arrowhead at the end of the ASK, and use the sender's variable as input to ASK.

ask Sprite(3) for list its variable with inputs my variable

In this example, "its variable" is defined in Sprite(3), so it can be used inside the expression. "My variable" is defined in the asking sprite, so it can't be used inside the expression. Instead it is used as input to ASK, which fills the empty slot in the LIST block with it.

0.112.1 Example Images

No examples yet.

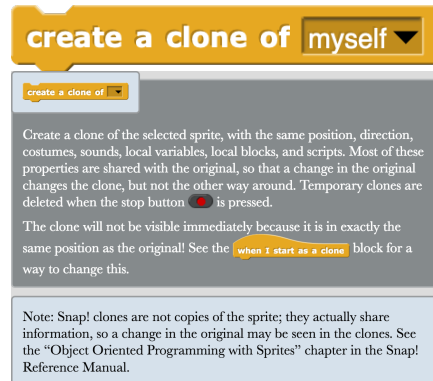
0.112.2 Example Projects

No examples yet.

113

0.113 create a clone of

Complete Me



0.113.1 Example Images

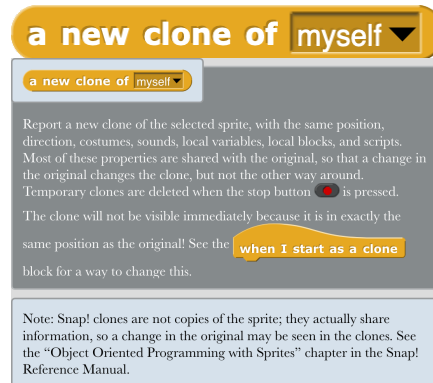
No examples yet.

0.113.2 Example Projects

No examples yet.

0.114 a new clone of

Complete Me



0.114.1 Example Images

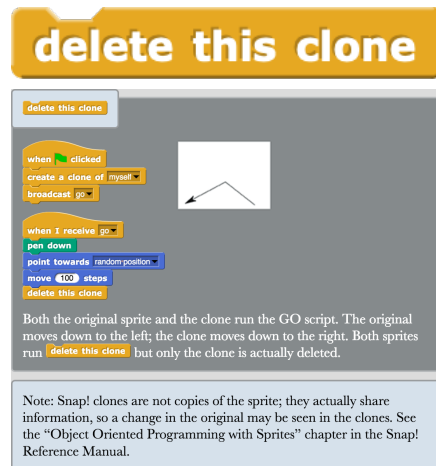
No examples yet.

0.114.2 Example Projects

No examples yet.

0.115 delete this clone

Complete Me



0.115.1 Example Images

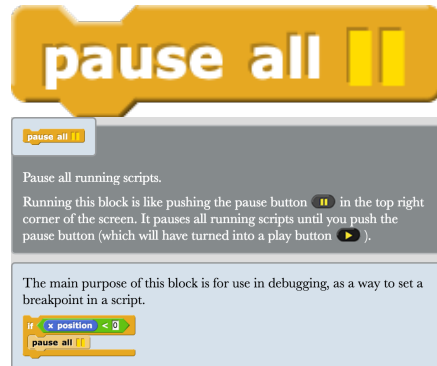
No examples yet.

0.115.2 Example Projects

No examples yet.

0.116 pause all ☒

Complete Me



0.116.1 Example Images

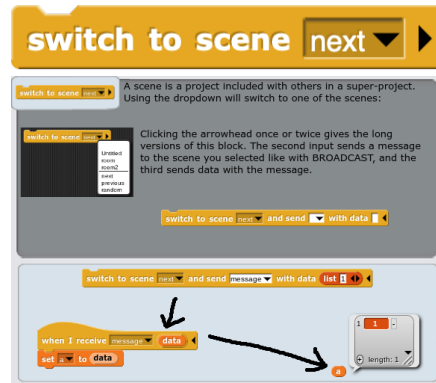
No examples yet.

0.116.2 Example Projects

No examples yet.

0.117 switch to scene

Complete Me



0.117.1 Example Images

No examples yet.

0.117.2 Example Projects

No examples yet.

0.118 define

Complete Me



0.118.1 Example Images

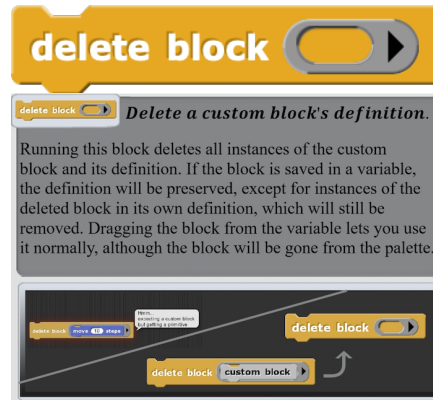
No examples yet.

0.118.2 Example Projects

No examples yet.

0.119 delete block

Complete Me



0.119.1 Example Images

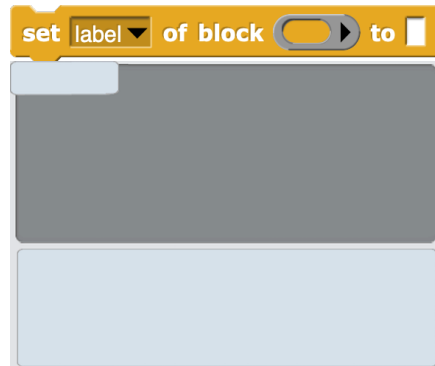
No examples yet.

0.119.2 Example Projects

No examples yet.

0.120 set of block to

Complete Me



0.120.1 Example Images

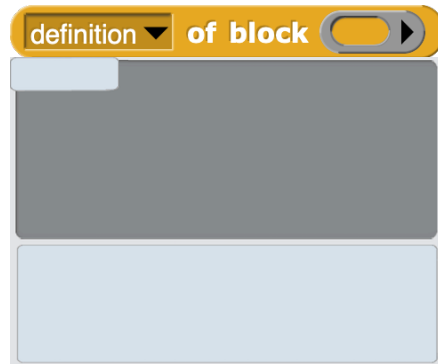
No examples yet.

0.120.2 Example Projects

No examples yet.

0.121 Attribute Of Block

Complete Me



0.121.1 Example Images

No examples yet.

0.121.2 Example Projects

No examples yet.

0.122 this

Complete Me



0.122.1 Example Images

No examples yet.

0.122.2 Example Projects

No examples yet.

0.123 set slot to

Complete Me



0.123.1 Example Images

No examples yet.

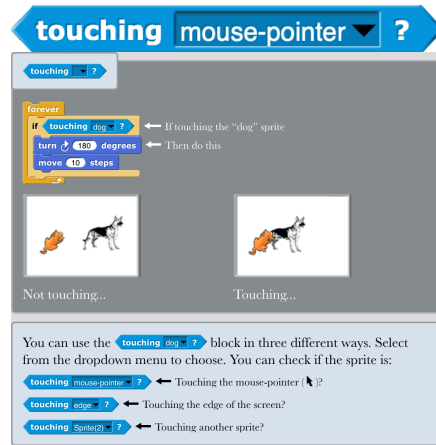
0.123.2 Example Projects

No examples yet.

0.124 Sensing Blocks

0.125 touching ?

Complete Me



0.125.1 Example Images

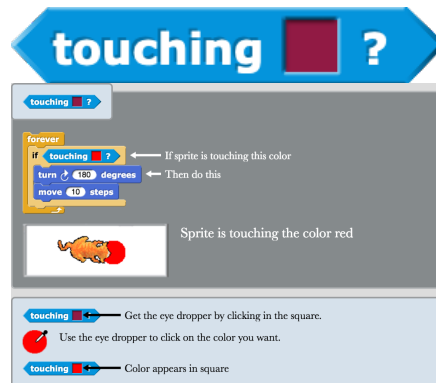
No examples yet.

0.125.2 Example Projects

No examples yet.

0.126 touching ?

Complete Me



0.126.1 Example Images

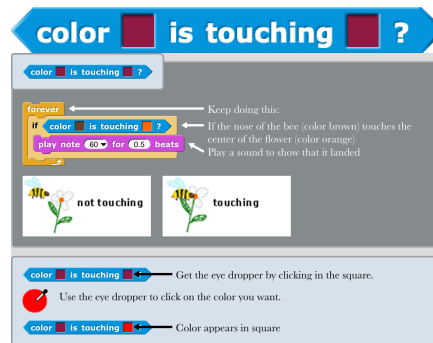
No examples yet.

0.126.2 Example Projects

No examples yet.

0.127 color is touching ?

Complete Me



0.127.1 Example Images

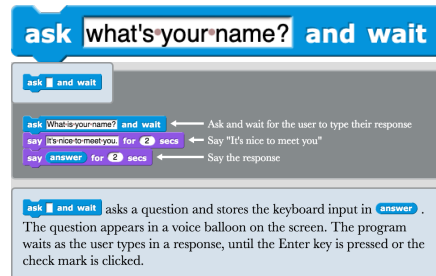
No examples yet.

0.127.2 Example Projects

No examples yet.

0.128 ask and wait

Complete Me



0.128.1 Example Images

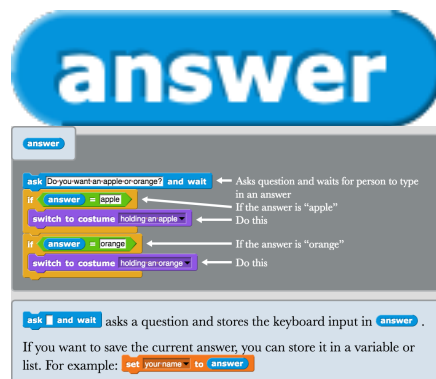
No examples yet.

0.128.2 Example Projects

No examples yet.

0.129 answer

Complete Me



0.129.1 Example Images

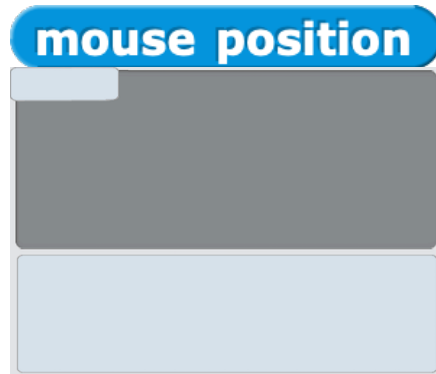
No examples yet.

0.129.2 Example Projects

No examples yet.

0.130 mouse position

Complete Me



0.130.1 Example Images

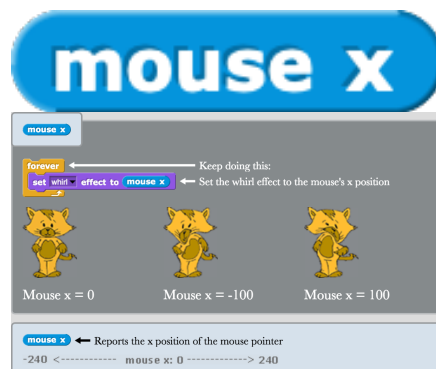
No examples yet.

0.130.2 Example Projects

No examples yet.

0.131 mouse x

Complete Me



0.131.1 Example Images

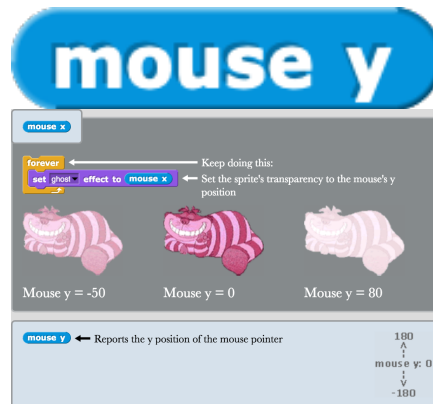
No examples yet.

0.131.2 Example Projects

No examples yet.

0.132 mouse y

Complete Me



0.132.1 Example Images

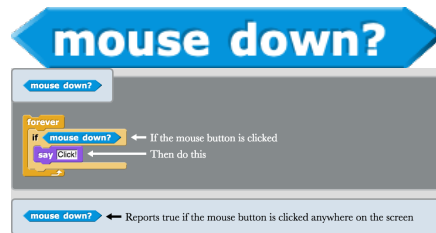
No examples yet.

0.132.2 Example Projects

No examples yet.

0.133 mouse down?

Complete Me



0.133.1 Example Images

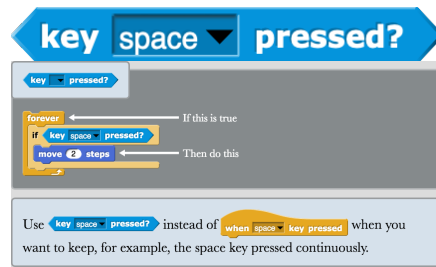
No examples yet.

0.133.2 Example Projects

No examples yet.

0.134 key pressed?

Complete Me



0.134.1 Example Images

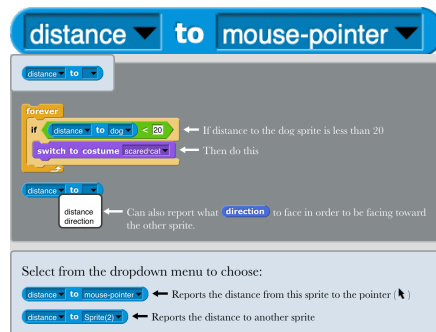
No examples yet.

0.134.2 Example Projects

No examples yet.

0.135 Distance To

Complete Me



0.135.1 Example Images

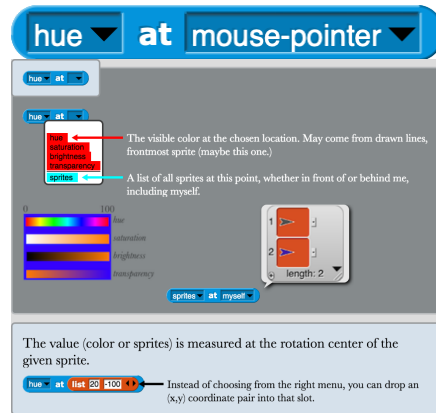
No examples yet.

0.135.2 Example Projects

No examples yet.

0.136 Color at Location

Complete Me



0.136.1 Example Images

No examples yet.

0.136.2 Example Projects

No examples yet.

0.137 reset timer

Complete Me



0.137.1 Example Images

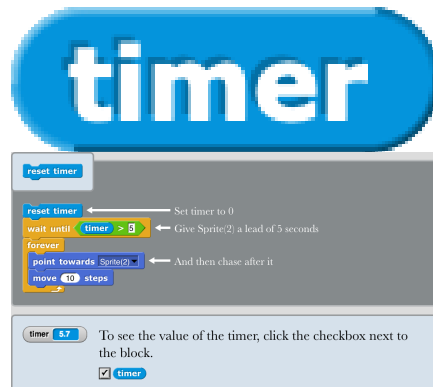
No examples yet.

0.137.2 Example Projects

No examples yet.

0.138 timer

Complete Me



0.138.1 Example Images

No examples yet.

0.138.2 Example Projects

No examples yet.

0.139 **current**

Complete Me

currentdate

currentdate

Report current date or time

year
month
date
day of week
hour
minute
seconds
time in milliseconds

Report current year, e.g., 2015

Report month of year, Jan=1, Dec=12

Report day number within month, 1-31

Report day of week (Sun=0, Mon=1, ... Sat=6)

Report hour of day, 0-23

Report minute within hour, 0-59

Report second within minute, 0-59

Report milliseconds since midnight, Jan 1, 1970

All reported values are numbers; you can write blocks to convert week days or months into words. Note that date numbers start at 1, as in conventional written notation (7/20/2015 or 20-7-2015 for July 20th), but time numbers start at 0, as in 24-hour clock notation (13:49:20 is 3:49pm plus 20 seconds).

joincurrentmonthcurrentdatecurrentyear

5/25/2015

0.139.1 **Example Images**

No examples yet.

0.139.2 **Example Projects**

No examples yet.

140

0.140 Attribute Of

Complete Me

costume# ▾ of ▾

costume# ▾ of ▾

Report an attribute of an object.

x position ▾ of Sprite2 ▾

(The attributes available in the left pulldown are different for different objects, e.g., the stage has no x position.)

Sprite2 my variable something ▾

If a sprite has a “for this sprite only” variable, it will appear in the left pulldown, so one sprite can examine the variables of another.

my variables ▾ of Sprite2 ▾

run

pen down ▾ of Sprite2 ▾

move 100 steps ▾ of Sprite2 ▾

By putting a script in a grey ring  and dragging it over the left pulldown, you can even let one sprite control the actions of another sprite!



0.140.1 Example Images

No examples yet.

0.140.2 Example Projects

No examples yet.

0.141 my

Complete Me



0.141.1 Example Images

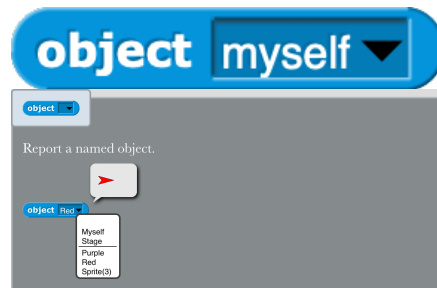
No examples yet.

0.141.2 Example Projects

No examples yet.

0.142 object

Complete Me



0.142.1 Example Images

No examples yet.

0.142.2 Example Projects

No examples yet.

0.143 url

Complete Me

url snap.berkeley.edu

Report web page text.

```
<html>
<head>
<meta name="snap-cloud-domain" location="https://snap.berkeley.edu:443">
<meta charset="UTF-8">
<title>Snap! Build Your Own Blocks</title>
<meta name="description" content="The Snap! social platform">
<meta name="author" content="Bernat Romagosa">
<meta name="snap-cloud-domain" location="https://snap.berkeley.edu/">
<link rel="icon" href="favicon.ico">
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Hint: Use SPLIT to see the result as a list of lines.

split url snap.berkeley.edu by line

Note: Many websites won't work with this block because of an Internet security feature built into browsers, called "Cross Origin Resource Sharing." There's nothing Snap! can do about this; you should try to get the provider of the website to allow cross-origin requests from everyone, or at least for your Internet domain. You can also look for a "CORS proxy," which is a website that funnels requests for arbitrary pages regardless of CORS status.

0.143.1 Example Images

No examples yet.

0.143.2 Example Projects

No examples yet.

0.144 microphone

Complete Me

microphone volume

Report data from microphone.
This block collects and analyzes a short (0.01 to 0.1 second) burst of sound.

microphone volume

volume

note

frequency

samples

spectrum

resolution

The first three options report a single number characterizing the sound as a whole average volume, dominant note number, and frequency.

The next two options are time domain information: a list of instantaneous volumes and a number representing the number of samples per second (sampling rate).

The last two options are frequency domain information: a histogram of frequencies and the number of buckets in the histogram.

“Time domain” means that the list items are the instantaneous volume at each moment, measured thousands of times per second. “Frequency domain” means that the list items are the volume over the entire burst of sound at each note frequency.

time domain



freq. domain



time domain



freq. domain



Tuning fork (pure sine wave)

Simple musical instrument with harmonics

0.144.1 Example Images

No examples yet.

0.144.2 Example Projects

No examples yet.

0.145 video on

Complete Me

video

motion

on

myself

video

motion

on

myself

Reports information from video capture.

The second input is a menu of sprites, and restricts the block's attention to the part of the stage taken up by the chosen sprite. ("Stage" uses the entire stage area.)

The first input specifies what information you want:

video

motion

on

myself

snapshot

motion

direction

Reports a snapshot of the video as a costume.

Reports a number, the speed of a detected moving object in the video.

Reports a number, the direction in which the object is moving, suitable for use as input in POINT IN DIRECTION.

0.145.1 Example Images

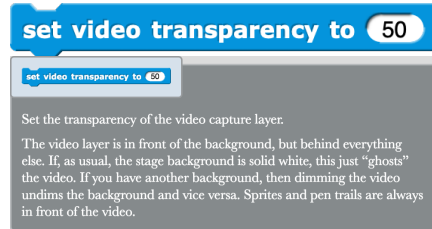
No examples yet.

0.145.2 Example Projects

No examples yet.

0.146 set video transparency to

Complete Me



0.146.1 Example Images

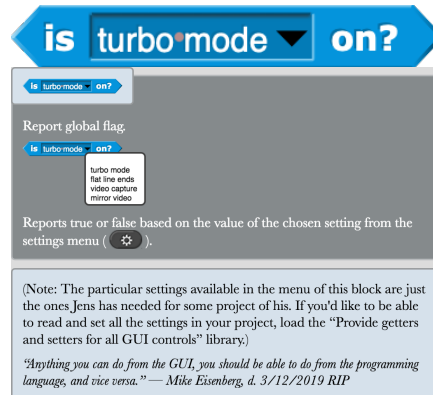
No examples yet.

0.146.2 Example Projects

No examples yet.

0.147 is on?

Complete Me



0.147.1 Example Images

No examples yet.

0.147.2 Example Projects

No examples yet.

0.148 set to

Complete Me



0.148.1 Example Images

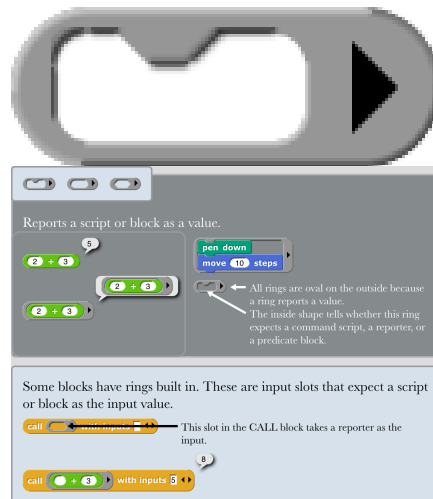
No examples yet.

0.148.2 Example Projects

No examples yet.

0.150 Command Ring

Complete Me



0.150.1 Example Images

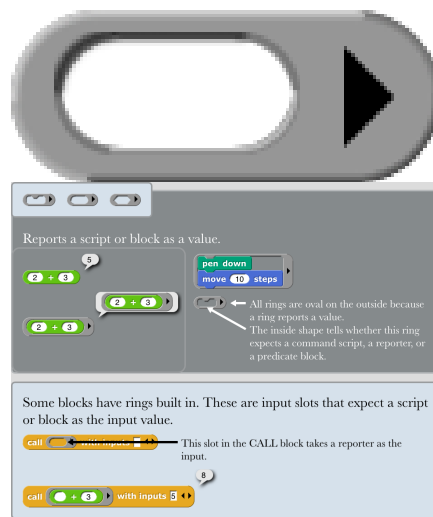
No examples yet.

0.150.2 Example Projects

No examples yet.

0.151 Reporter Ring

Complete Me



0.151.1 Example Images

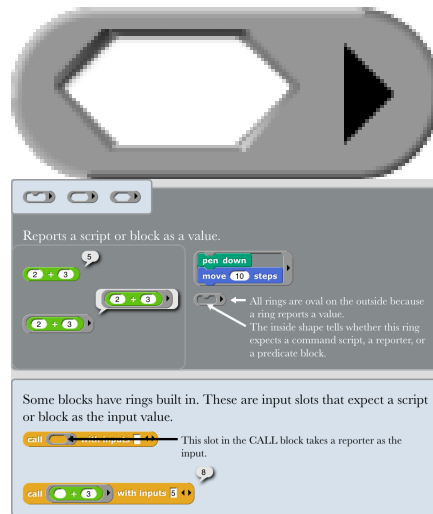
No examples yet.

0.151.2 Example Projects

No examples yet.

0.152 Predicate Ring

Complete Me



0.152.1 Example Images

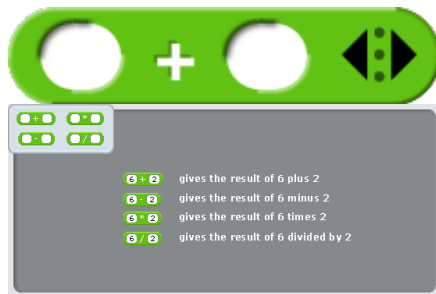
No examples yet.

0.152.2 Example Projects

No examples yet.

0.153 Sum +

Complete Me



0.153.1 Example Images

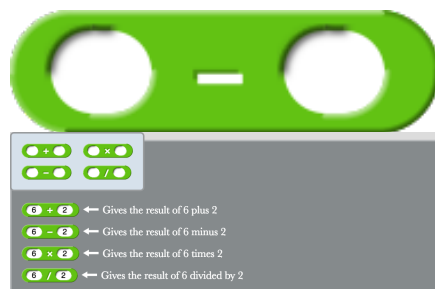
No examples yet.

0.153.2 Example Projects

No examples yet.

0.154 Subtract -

Complete Me



0.154.1 Example Images

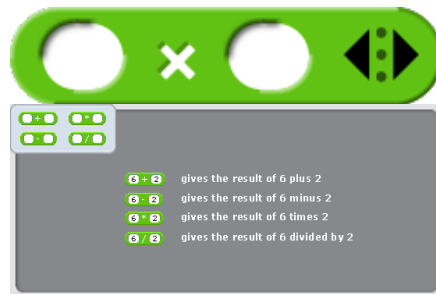
No examples yet.

0.154.2 Example Projects

No examples yet.

0.155 Product ×

Complete Me



0.155.1 Example Images

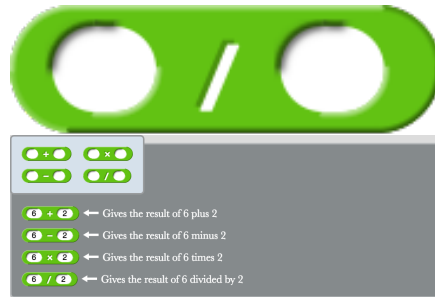
No examples yet.

0.155.2 Example Projects

No examples yet.

0.156 Divide ÷

Complete Me



0.156.1 Example Images

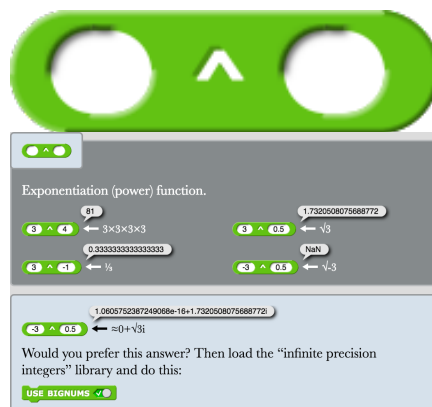
No examples yet.

0.156.2 Example Projects

No examples yet.

0.157 Power of Number

Complete Me



0.157.1 Example Images

No examples yet.

0.157.2 Example Projects

No examples yet.

0.158 Mod

Complete Me

mod

mod

forever

move 1 steps

if x position mod 10 = 0

change y by 10

wait 0.3 secs

change y by 10

Keep doing this

Take one step

Every ten steps

Jump

mod gives the remainder of the division of the two numbers.

For example, 143 mod 6 is 5 (143 divided 6 is 23 and the remainder is 5).

0.158.1 Example Images

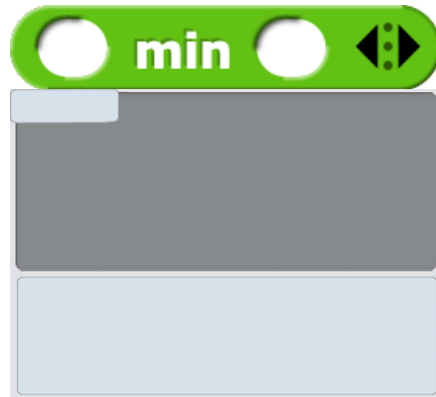
No examples yet.

0.158.2 Example Projects

No examples yet.

0.159 min

Complete Me



0.159.1 Example Images

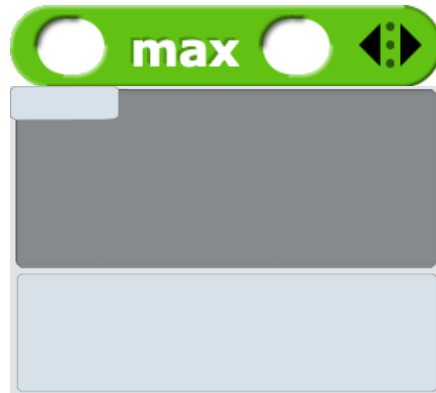
No examples yet.

0.159.2 Example Projects

No examples yet.

0.160 max

Complete Me



0.160.1 Example Images

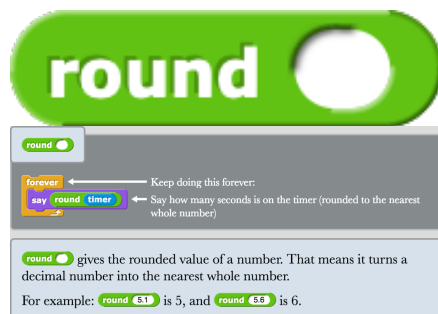
No examples yet.

0.160.2 Example Projects

No examples yet.

0.161 round

Complete Me



0.161.1 Example Images

No examples yet.

0.161.2 Example Projects

No examples yet.

0.162 Math Functions

Complete Me

sqrt ▾ of 10

sqrt ▾ of

pen down

forever

change x by 1

set y to 100 x sqrt of x position

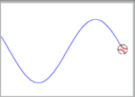
Put the pen down

Forever

Move to the right

Set the y position to a function of the x position

Set y based on the sine of x



sqrt ▾ of 10

abs

neg

ceiling

floor

sqrt

sin

cos

tan

asin

acos

atan

ln

log

10

e^x

10^x

2^x

This block performs a mathematical function. Choose the math function from the block's drop down menu.

Note: Trig functions are based on angles in degrees, not radians!

0.162.1 Example Images

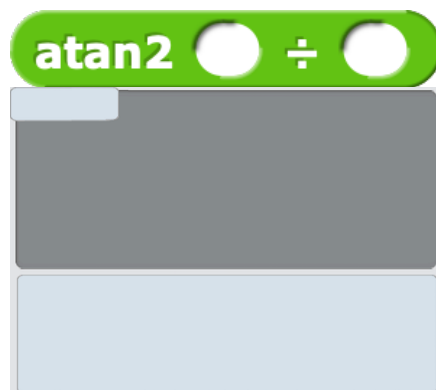
No examples yet.

0.162.2 Example Projects

No examples yet.

0.163 atan2 ÷

Complete Me



0.163.1 Example Images

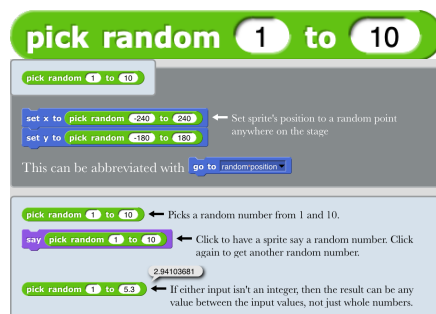
No examples yet.

0.163.2 Example Projects

No examples yet.

0.164 pick random to

Complete Me



0.164.1 Example Images

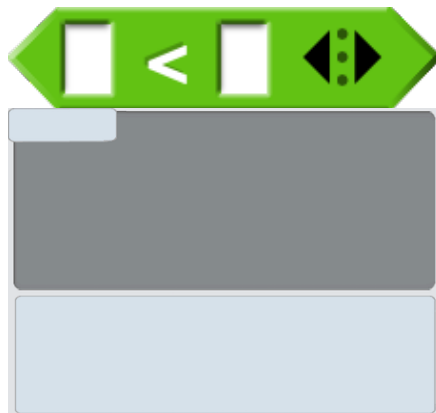
No examples yet.

0.164.2 Example Projects

No examples yet.

0.165 <

Complete Me



0.165.1 Example Images

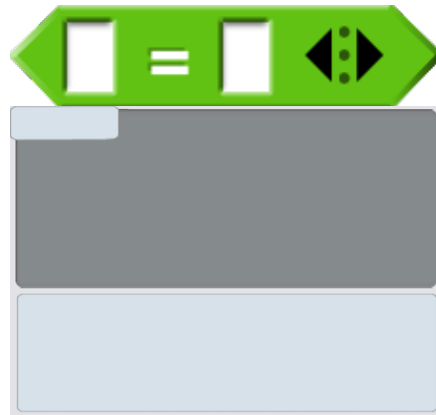
No examples yet.

0.165.2 Example Projects

No examples yet.

0.166 _

Complete Me



0.166.1 Example Images

No examples yet.

0.166.2 Example Projects

No examples yet.

0.167 >

Complete Me



0.167.1 Example Images

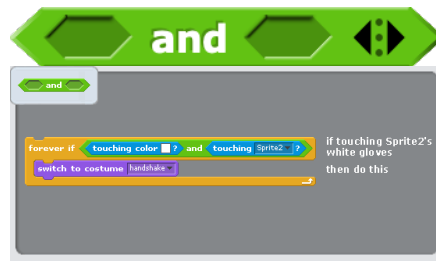
No examples yet.

0.167.2 Example Projects

No examples yet.

0.168 _

Complete Me



0.168.1 Example Images

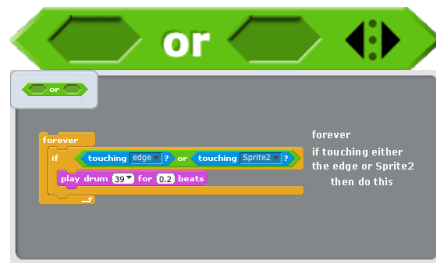
No examples yet.

0.168.2 Example Projects

No examples yet.

0.169 or

Complete Me



0.169.1 Example Images

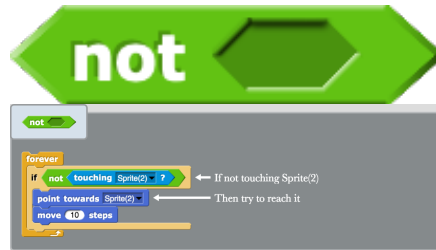
No examples yet.

0.169.2 Example Projects

No examples yet.

0.170 not

Complete Me



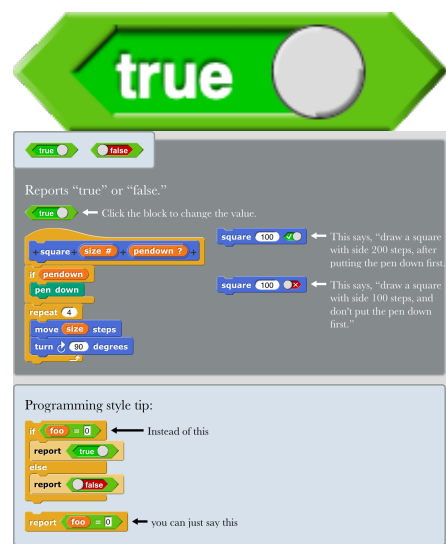
0.170.1 Example Images

No examples yet.

0.170.2 Example Projects

No examples yet.

Complete Me



0.171.1 Example Images

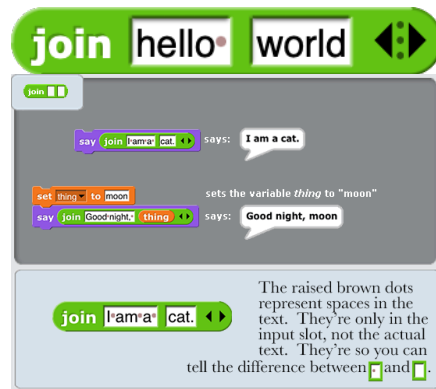
No examples yet.

0.171.2 Example Projects

No examples yet.

0.172 join

Complete Me



0.172.1 Example Images

No examples yet.

0.172.2 Example Projects

No examples yet.

0.173 split by

Complete Me

split

hello-world

by

split hello-world by

Report list of text split into words or lines.

1 nowhere

2

3

4

5

6

7

length: 4

split hello-world by

Splitting on a character (space, in this example) can result in empty list elements if the character appears several times in a row in the text.

1 nowhere

2

3

4

5

6

7

length: 2

split nowhere-man by whitespace

Choosing "whitespace" from the pulldown menu will treat any number of spaces, tabs, or newlines as a single separator, to make a list of words.

split Yaksio by letter

Split a text into individual characters (including spaces, newlines, and punctuation as list items).

332 items

1 <html>

2 <head>

3 <meta name="snap-o

4 <meta charset="L

5 <title>Snap! Ban

6 <meta name="des

7 <meta name="s

split url snap.berkeley.edu by line

Split a multiline text into lines. (The newline character, the carriage return character, or the combination CR, NL all count as a single line break in the text.)

1 b

2

3

4

5

6

7

length: 3

split banana by an

It's not often useful, but you can use a multi-character string as the separator. In this example, we get an empty list item because "an" appears twice in a row in "banana."

0.173.1 Example Images

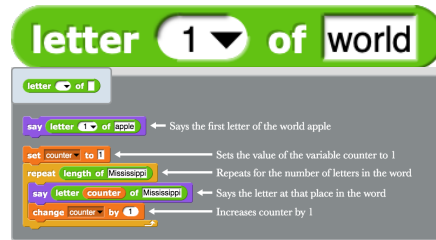
No examples yet.

0.173.2 Example Projects

No examples yet.

0.174 letter of

Complete Me



0.174.1 Example Images

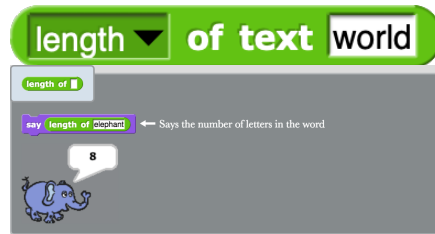
No examples yet.

0.174.2 Example Projects

No examples yet.

0.175 Attribute of Text

Complete Me



0.175.1 Example Images

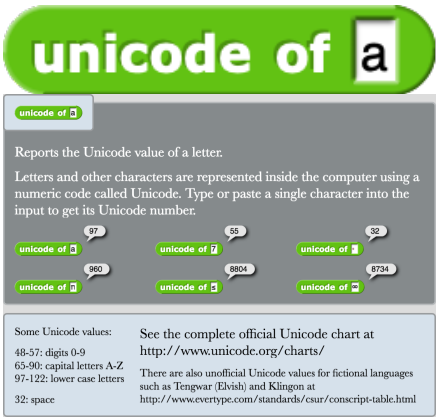
No examples yet.

0.175.2 Example Projects

No examples yet.

0.176 **unicode of**

Complete Me



0.176.1 **Example Images**

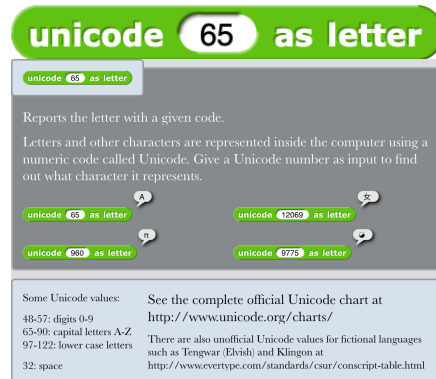
No examples yet.

0.176.2 **Example Projects**

No examples yet.

0.177 unicode as letter

Complete Me



0.177.1 Example Images

No examples yet.

0.177.2 Example Projects

No examples yet.

0.178 is a ?

Complete Me

is 5 a number ?

is 5 a number ?

Reports **true** if input is a given type.

All of these are true:

is 537 a number ?

is 5 a text ?

is 3 + 4 a number ?

is true a text ?

is 3 + 4 a reporter ?

is true a Boolean ?

is pen up a command ?

is true a predicate ?

The possible types are number, text, Boolean (true or false), list, sprite, costume, sound (object-oriented types), command (a script or jigsaw-shaped command block), reporter (an oval block that reports a value), and predicate (a hexagonal block that reports true or false). Every value has exactly one of these types.

is 5 a number ?

number
text
Boolean
list
sprite
costume
sound
command
reporter
predicate

0.178.1 Example Images

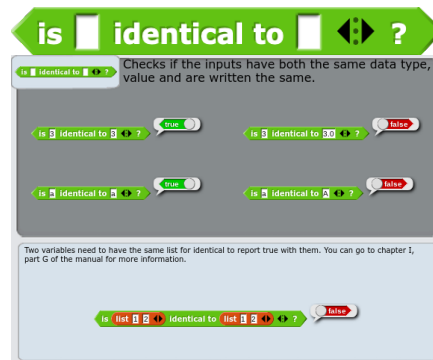
No examples yet.

0.178.2 Example Projects

No examples yet.

0.179 is ?

Complete Me



0.179.1 Example Images

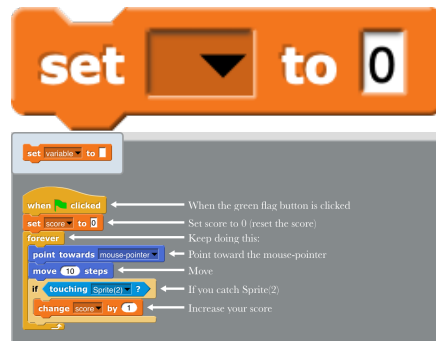
No examples yet.

0.179.2 Example Projects

No examples yet.

0.181 set to

Complete Me



0.181.1 Example Images

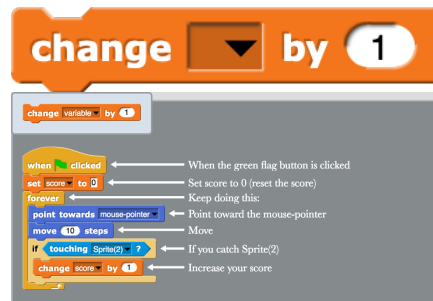
No examples yet.

0.181.2 Example Projects

No examples yet.

0.182 change by

Complete Me



0.182.1 Example Images

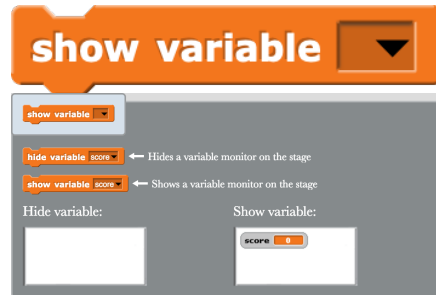
No examples yet.

0.182.2 Example Projects

No examples yet.

0.183 show variable

Complete Me



0.183.1 Example Images

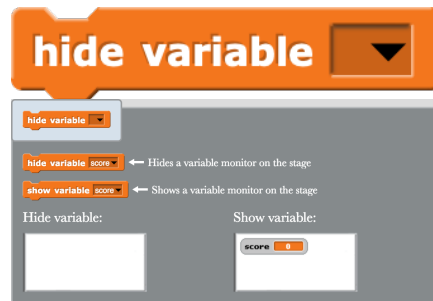
No examples yet.

0.183.2 Example Projects

No examples yet.

0.184 hide variable

Complete Me



0.184.1 Example Images

No examples yet.

0.184.2 Example Projects

No examples yet.

0.185 Script variables

Complete Me

script variables **a**

script variables **a**

swap items index1 # and index2 # in list

script variables temp

set temp to item index1 of list

replace item index1 of list with item index2 of list

replace item index2 of list with temp

Every time this block is called, it makes a new temporary variable **temp**. Click (don't drag) to change name.

Script variables don't clutter up your palette, and you can be sure no other block changes their value. This works for scripting arca scripts, too, not just custom blocks.

new counter

script variables count

set count to 0

report change count by 1

report report count

Object Oriented Programming:
If a reporter reports a script, and that script uses a script variable created outside the reported script (such as the use of **count** in the script that **new counter** reports) then the script variable isn't temporary!

set mycounter to new counter

say call my counter for 2 secs → 1

say call my counter for 2 secs → 2

0.185.1 Example Images

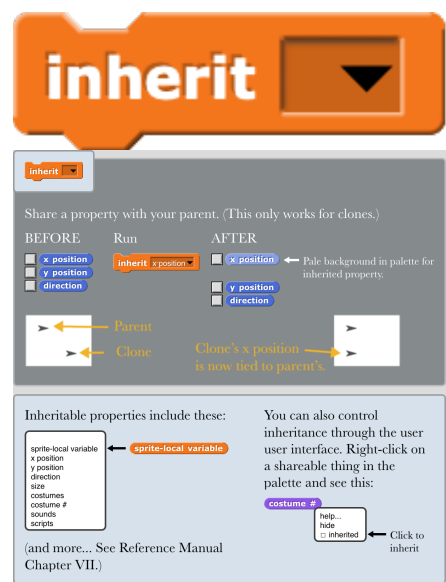
No examples yet.

0.185.2 Example Projects

No examples yet.

0.186 inherit

Complete Me



0.186.1 Example Images

No examples yet.

0.186.2 Example Projects

No examples yet.

0.187 list

Complete Me



0.187.1 Example Images

No examples yet.

0.187.2 Example Projects

No examples yet.

0.188 numbers from to

Complete Me

numbers from 1 to 10

numbers from 1 to 10

Report a list of consecutive numbers.

1

2

3

4

length: 4

1

2

3

4

length: 4

numbers from 3 to 3

numbers from 2 to 1

length: 0

numbers from 3 to 1

(Range must be ascending.)

prime? value #

keep items

value mod 2 = 0

from

empty?

report is

numbers from 2 to value = 1

A number n is prime if the list of its factors in the range [2,n-1] is empty.

prime? 2

prime? 3

1

2

3

4

length: 4

numbers from 1.5 to 5

0.188.1 Example Images

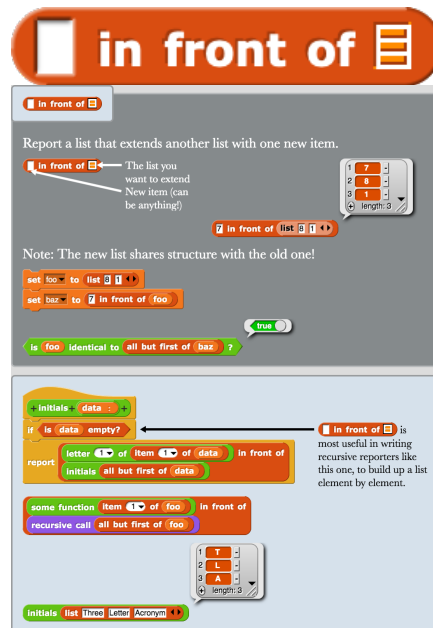
No examples yet.

0.188.2 Example Projects

No examples yet.

0.189 In Front Of

Complete Me



0.189.1 Example Images

No examples yet.

0.189.2 Example Projects

No examples yet.

0.190 item of

Complete Me

item1 of

Item 1 of

say item random of phrases

← Say one of the items from a list of phrases

if score > item 1 of top scores

← If the current score is greater than item 1 on a top score list

play sound clap

→ Play a clapping sound

The item block reports the value of the item at the specified place on a list.

Item 1 of

Select from the menu or insert a number to indicate which item you want

1
last
random

You can fit an item block into other blocks, for example: say, switch to costume, play sound, or broadcast.

0.190.1 Example Images

No examples yet.

0.190.2 Example Projects

No examples yet.

0.191 all but first of

Complete Me

all but first of

all but first of

Report all but first list item.

18

21

length: 2

all but first of list 7 3 1

The new list shares structure with the old one; it's not a copy. If the input list was made with the list reporter blocks **In front of** **Item** **of** **all but first of** then **all but first of** is very fast; it doesn't have to copy the items or the list structure. If you use list commands such as **Insert item at** **of** on a list, then it will be a a dynamic array rather than a linked list, so this block has to convert to linked form.

The list reporters are good for writing recursive list reporters that build up a result list item by item.

0.191.1 Example Images

No examples yet.

0.191.2 Example Projects

No examples yet.

0.192 Report List Attribute

Complete Me

length of

length of

length of list 3 3 3 4

Reports an attribute of the list

length of list 3 list 3 4 4

It gives the number of items in the *automat* list.

length of list 3 list 3 4 4

Here are the possible attributes:

length
rank
arrays one
flatten
columns
reverse

LENGTH is described above
RANK reports the LENGTH OF DIMENSIONS OF its input
DIMENSIONS reports how long the list is in each dimension
FLATTEN reports each non-list item, even those from sublists.
E.g., FLATTEN OF (a (b c) d e f (g h)) gives (a b c d e f g)
COLLIMNS, given (a b c) (d e f), reports (a d) (b e) (c f)
REVERSE, given (a b (c d e) i), reports f (e d c) b a)

These three report texts, not lists:
LINPS joins each item with a newline
CSV takes a 2D list and gives the spreadsheet equivalent
JSON takes an arbitrarily deep list, often with key-value
pairs, and gives the JSON equivalent

array
list
not

0.192.1 Example Images

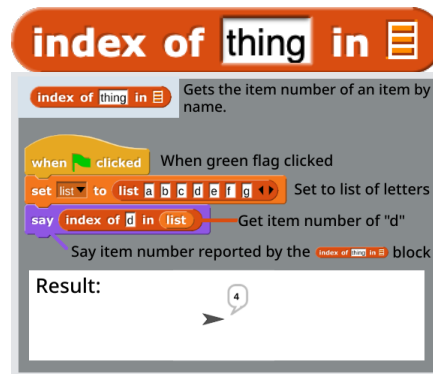
No examples yet.

0.192.2 Example Projects

No examples yet.

0.193 index of in

Complete Me



0.193.1 Example Images

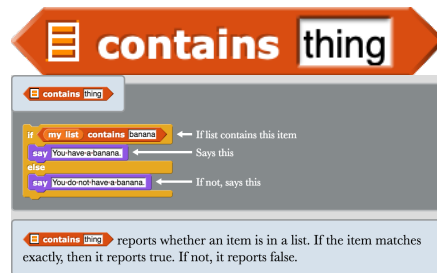
No examples yet.

0.193.2 Example Projects

No examples yet.

0.194 List Contains

Complete Me



0.194.1 Example Images

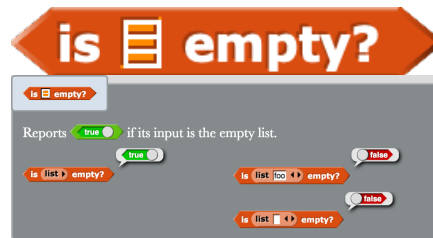
No examples yet.

0.194.2 Example Projects

No examples yet.

0.195 is empty?

Complete Me



0.195.1 Example Images

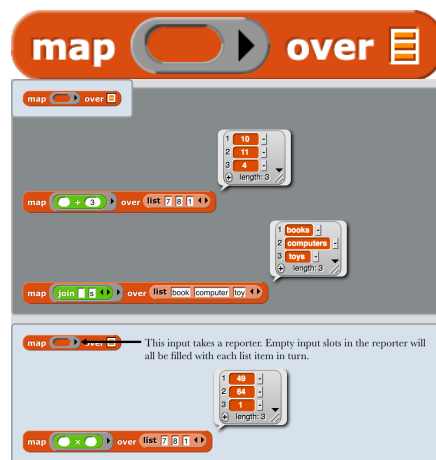
No examples yet.

0.195.2 Example Projects

No examples yet.

0.196 map over

Complete Me



0.196.1 Example Images

No examples yet.

0.196.2 Example Projects

No examples yet.

0.197 keep items from

Complete Me



The image shows the 'keep items from' block in a Scratch-like environment. The block has a title bar with 'keep items' and 'from' and a list icon. Below the title bar, there are two examples of the block being used. The first example shows the block with a 'number > 2' predicate and a list containing 'cat', 'dog', 'fish', 'hen', 'turtle', and 'dove'. The second example shows the block with a 'length of > 3' predicate and a list containing 'University of California at Berkeley'. Below the examples, there is a text box explaining the block's functionality.

Each list item is put into the empty input slot of the predicate block. If the predicate reports **true** then that item is included in the result list. Note that if the predicate has a default input value, you have to clear the slot to use it with KEEP; e.g., this won't work:

KEEP makes a new list; the original list is not modified.

0.197.1 Example Images

No examples yet.

0.197.2 Example Projects

No examples yet.

0.198 find first item in

Complete Me

find first item  **in** 

find first item  **in** 

Reports the first item in the list that matches the predicate.

find first item    **in** **list**       

Compare:

1	University
2	California
3	Berkeley
length	3

keep items    **from** **list**       

find first item    **in** **list**       

False

keep items    **from** **list**       

FIND FIRST reports False if no item in the list matches the predicate. (KEEP reports the empty list in that situation.)

FIND FIRST means the same as **item**  **of** **keep items**  **from**  but FIND FIRST is faster if a match is found because it doesn't search the rest of the list.

0.198.1 Example Images

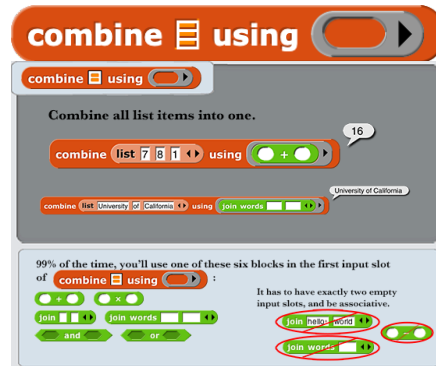
No examples yet.

0.198.2 Example Projects

No examples yet.

0.199 combine using

Complete Me



0.199.1 Example Images

No examples yet.

0.199.2 Example Projects

No examples yet.

0.200 for each in

Complete Me



0.200.1 Example Images

No examples yet.

0.200.2 Example Projects

No examples yet.

0.201 add to

Complete Me



0.201.1 Example Images

No examples yet.

0.201.2 Example Projects

No examples yet.

0.202 delete of

Complete Me



0.202.1 Example Images

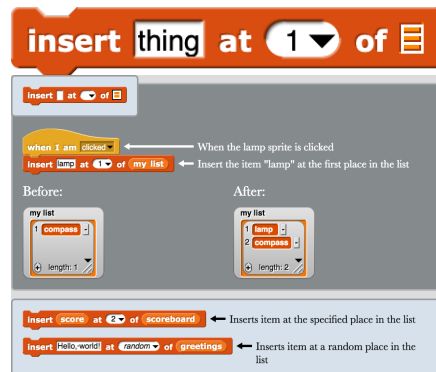
No examples yet.

0.202.2 Example Projects

No examples yet.

0.203 insert at of

Complete Me



0.203.1 Example Images

No examples yet.

0.203.2 Example Projects

No examples yet.

0.204 **replace item of with**

Complete Me

replace item 1 of with thing

replace item of with

replace item 1 of scoreboard with score ← Replace item 1 in the list with the current score

Before:

After:

scoreboard

1 3 2 8 length: 2

scoreboard

1 10 2 8 length: 2

You can choose where in a list to put an item. For example:

replace item 2 of race times with timer ← Replaces item 2 in the list

replace item 4 of questions with How are you? ← Replaces last item in the list

0.204.1 **Example Images**

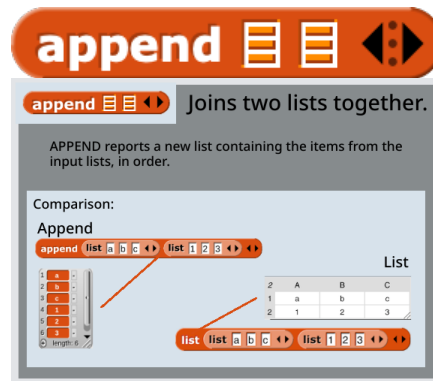
No examples yet.

0.204.2 **Example Projects**

No examples yet.

0.205 append

Complete Me



0.205.1 Example Images

No examples yet.

0.205.2 Example Projects

No examples yet.

0.206 reshape to

Complete Me

reshape to 4 3

Reshapes a list to the provided dimensions

Examples:

reshape numbers from 1 to 10 to 5 2

5	A	B
1	1	2
2	3	4
3	5	6
4	7	8
5	9	10

If the dimensions multiply to a number larger than the length of the list, it starts over with the beginning of the list:

reshape numbers from 1 to 10 to 4 3

4	A	B	C	D
1	1	2	3	4
2	5	6	7	8
3	9	10	1	2
4	3	4	5	6

You can use a simple scalar input (number or text string) too:

reshape join 5 to 4 3

4	A	B	C
1	5	5	5
2	5	5	5
3	5	5	5
4	5	5	5

And with no dimensions, it returns a scalar:

reshape numbers from 1 to 10 to

0.206.1 Example Images

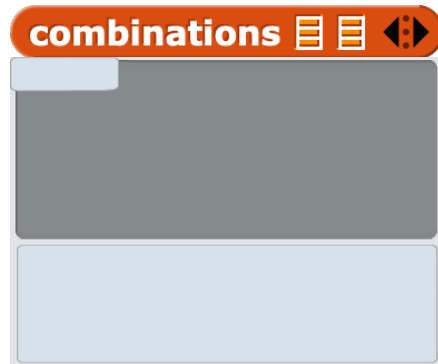
No examples yet.

0.206.2 Example Projects

No examples yet.

0.207 combinations

Complete Me



0.207.1 Example Images

No examples yet.

0.207.2 Example Projects

No examples yet.